研究文章 基于多器官毒理基因组学数据预测药物诱导病理的多标签学习模型

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数据可用性声明：用于进行实验、模型拟合和绘图的所有数据和代码可在 GitHub 仓库 https://github.com/RanSuLab/Drug-Toxicity-Prediction-MultiLabel 获取。

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摘要

药物诱导的毒性会损害健康，是导致药物从市场撤回的关键因素之一。识别药物诱导的靶器官毒性具有重要意义，尤其是详细的病理学发现，这对在药物开发早期阶段进行毒性评估至关重要。大量研究致力于识别药物毒性。然而，大多数研究仅限于单一器官或仅涉及二元毒性。在此，我们提出了一种新型多标签学习模型，命名为 Att-RethinkNet，用于基于毒理基因组学数据预测针对肝脏和肾脏的药物诱导病理学发现。Att-RethinkNet 配备了记忆结构，能够有效利用标签关联信息。此外，嵌入注意力机制以关注重要特征并获得更好的特征表现。我们的 Att-RethinkNet 可应用于多个器官，并考虑化合物类型、剂量和给药方式。

作者摘要

药物诱导的毒性会损害健康，是导致药物撤市的关键因素之一。因此，为了全面评估药物诱导的毒性，预测详细的病理发现非常重要，这对于理解毒性机制也至关重要。然而，目前大多数现有的毒性研究只预测二元毒性（有毒或无毒）或仅预测单一器官的毒性。多器官的病理发现尚未得到充分探讨。本文通过提出的 Att-RethinkNet 展示了预测药物诱导病理发现的可能性。

RESEARCH ARTICLE

A multi-label learning model for predicting drug-induced pathology in multi-organ based on toxicogenomics data

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Abstract

Drug-induced toxicity damages the health and is one of the key factors causing drug withdrawal from the market. It is of great significance to identify drug-induced target-organ toxicity, especially the detailed pathological findings, which are crucial for toxicity assessment, in the early stage of drug development process. A large variety of studies have devoted to identify drug toxicity. However, most of them are limited to single organ or only binary toxicity. Here we proposed a novel multi-label learning model named Att-RethinkNet, for predicting drug-induced pathological findings targeted on liver and kidney based on toxicogenomics data. The Att-RethinkNet is equipped with a memory structure and can effectively use the label association information. Besides, attention mechanism is embedded to focus on the important features and obtain better feature presentation. Our Att-RethinkNet is applicable in multiple organs and takes account the compound type, dose, and administration time, so it is more comprehensive and generalized. And more importantly, it predicts multiple pathological findings at the same time, instead of predicting each pathology separately as the previous model did. To demonstrate the effectiveness of the proposed model, we compared the proposed method with a series of state-of-the-arts methods. Our model shows competitive performance and can predict potential hepatotoxicity and nephrotoxicity in a more accurate and reliable way. The implementation of the proposed method is available at https://github.com/RanSuLab/Drug-Toxicity-Prediction-MultiLabel.

Author summary

Drug-induced toxicity damages the health and is one of the key factors causing drug withdrawal from the market. Hence, to fully assess drug-induced toxicity, it is important to predict the detailed pathological findings, which are also crucial for toxicity mechanism understanding. However, most of the existing toxicity studies only predict binary toxicity (the toxicity or non-toxicity) or only predict the toxicity targeting single organ. The pathological findings of multiple organs are not well explored. Here we show, through the proposed Att-RethinkNet, it is possible to predict drug-induced pathological findings on

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肝脏和肾脏。我们的结果表明，Att-RethinkNet 能以更准确可靠的方式预测潜在的肝毒性和肾毒性，而且它适用于多种器官，并考虑化合物类型、剂量和给药时间，因此比现有方法更全面、更通用。在多器官上准确预测病理发现可能有利于药物开发。

这是一篇 *PLOS* 计算生物学方法论文。

介绍

药物开发包括将新药从实验室带到市场的相关活动。它通常分为四个不同且必不可少的阶段：药物发现、临床前研究、临床研究以及批准和营销，这一过程相对昂贵且耗时，并充满风险和不确定性。通过对药物开发成本相关统计数据的系统回顾，研究人员发现，公司通常需要花费10到15年和数百万美元才能将新药推向市场 [1, 2]。然而，由于多种原因，新药候选者的失败率仍然相当高，而药物引起的毒性，包括不良反应和毒性作用，是药物被撤回或停用的常见原因 [3]。药物引起的毒性是通过对表型终点的病理学发现来评估的，指的是药物的不良影响，即由药物相互作用引起的功能障碍和组织损伤。

在过去，通过湿实验预测药物诱导的毒性是很常见的。虽然这种类型的实验不可替代，但它需要专门设计的实验室、安全设备、专业研究人员等，这是一种成本高且不方便的程序。因此，越来越多的研究人员对计算机模拟技术感兴趣，因为计算方法通常具有成本效益，这为开发新药提供了指导，并帮助研究人员在药物开发过程中评估药物的安全风险。近年来，在毒理学中，微阵列技术（称为毒理基因组学）正成为确定新化学实体潜在毒性的一种广泛使用的方法 [6–8]。毒理基因组学数据在理解和预测药物诱导的毒性方面起着重要作用，并且基因表达数据的应用促使研究人员通过数据分析方法解决生物学问题。基因表达分析

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both liver and kidney. Our results suggest that the Att-RethinkNet predicts potential hepatotoxicity and nephrotoxicity in a more accurate and reliable way, and it is applicable in multiple organs and takes account the compound type, dose, and administration time, so it is more comprehensive and generalized than the existing methods. The accurate prediction of pathological findings on multiple organs may benefit drug development.

This is a *PLOS Computational Biology* Methods paper.

Introduction

Drug development consists of activities involving in bringing a new drug from laboratory to market. It is typically divided into four distinct and essential phases: drug discovery, preclinical research, clinical research, and approval and marketing, which is relatively expensive and time-consuming, and is filled with risk and uncertainty. Through systematic review of statistics concerning the cost of drug development, researchers find that companies spend 10 to 15 years and millions of dollars in obtaining a new drug into the market [1, 2]. However, failure rate of new drug candidates is still considerably high for many reasons, and drug-induced toxicity, including adverse reactions and toxic effects, is a common reason for drug withdrawal or discontinuation [3]. Drug-induced toxicity, which is assessed by the pathological findings with respect to the phenotypic end point, refers to the negative effects of medications, that is, dysfunctions and tissue lesions caused by the interaction of various chemical substances, which may cause adverse health issues. Since kidney and liver are filters for various regions of the body, they are the primary targets of toxins [4] and reports have shown that a great deal of drug failure is due to the hepatotoxicity and nephrotoxicity [5]. Thus, it is necessary to identify drug-induced hepatotoxicity/nephrotoxicity, especially the pathological findings caused by hepatotoxicity/nephrotoxicity in the early stage of drug development and eliminate toxic compounds as soon as possible so that the success rate of drug candidate trials can be greatly improved.

In the past, it is common to predict drug-induced toxicity through wet-lab experiments. Although such type of experiment is irreplaceable, it requires specially designed room, safety equipment, professional researchers, etc., which is a costly and inconvenient procedure. Therefore, a growing number of researchers are interested in in-silico techniques because computational approaches are usually cost-effective, which provides guidance for developing a new pharmaceutical drug and assists researchers assessing drug safety risks during drug development. In recent years, micro-array technology in toxicology, known as toxicogenomics, is becoming a broadly used method for determination of potential toxicity of a new chemical entity [6–8]. Toxicogenomics data plays an important role in understanding and predicting drug-induced toxicity, and the application of gene expression data prompts researchers to solve biological problems through data analysis methods. The analysis of gene expression profiles in target organs after drug treatment can be used to assist in detecting potential toxicity before the appearance of a toxic phenotype [9–13]. Presently, some databases such as Open TG-GATEs (Toxicogenomics Project-Genomics Assisted Toxicity Evaluation System), which is one of the largest public toxicogenomics databases have been built for toxicity research [14]. And an increasing number of studies have focused on using the gene expression profiles and toxicity information for the identification of the potential negative effects of drugs.

许多研究人员已经对目标器官进行了系列毒性探索。Zhu 等人基于三种类型的描述符构建了随机森林模型，用于预测药物诱导的肝损伤（DILI），基于欠采样和过采样处理不平衡数据集的模型都取得了令人满意的表现 [15]。Minowa 等人提出了一种基于基因表达谱的预测模型，用于预测大鼠药物诱导的近曲小管损伤，并发现单次给药后 24 小时有大量基因表现出差异表达，从而在一定程度上提高了模型的预测能力 [16]。在此基础上，An 等人尝试同时考虑两个器官的毒性，并开发了四种计算模型来分类药物是肝毒性还是肝-肾毒性。模型使用了人工神经网络（ANN）、k 近邻（kNN）、线性判别分析（LDA）和支持向量机（SVM）。

在对最近的研究进行回顾后，我们得出结论：尽管一些研究取得了很高的准确率表现，但现有的工作仍然存在局限性。首先，大多数先前的研究集中在特定器官的毒性（有毒或无毒）预测上。考虑到其在毒性评估中的重要性，病理发现预测尚未被广泛探索。现有的少量病理预测模型为每个标签开发了独立模型，这忽略了化合物可能同时引起多种毒性效应的事实。其次，病理发现预测是一个多标签分类任务。近年来，在生物信息领域提出了许多多标签分类方法。然而，毒理基因组学领域的大多数现有多标签分类模型仍然使用传统的机器学习模型，如二元相关（BR）或分类器链（CC）。先进的深度学习技术已经不

在本文中，我们提出了一种新型的多标签学习模型，命名为 Att-RethinkNet，用于基于毒理基因组学数据预测药物对多器官的毒性。与处理区分化合物是否有毒或无毒的二分类问题不同，我们识别了肝脏和肾脏的特定病理发现，这是一个多标签学习任务。为了克服传统多标签分类中忽略标签相关性等缺点，受 Yang 等人的工作 [25] 启发——该工作在多个多标签数据集上进行了评估，并证明 RethinkNet 在多标签分类任务中比最先进的算法表现更优——我们设计了该深度框架。

Many researchers have carried out a series of toxicity exploration on target organs. Zhu et al. constructed random forest models based on three types of descriptors to predict drug-induced liver injury (DILI), the models based on under-sampling and over-sampling to deal with unbalanced data sets both achieved promising performance [15]. Minowa et al. proposed a prediction model based on gene expression profiles for predicting drug-induced proximal tubular injury in rats, and found that there were differentially expressed in a number of genes at 24h after a single dose administration, which improved the predictive powder of the model to a certain extent [16]. On this basis, An et al. tried to consider the toxicity of two organs at the same time, and developed four computational models to classify whether a drug is liver toxic or liver-kidney toxic. The models used artificial neural network (ANN), k-nearest neighbor (kNN), linear discriminant analysis (LDA), and support vector machine (SVM) respectively, and all prediction accuracy of them were more than 90% [17]. Zhang et al. mapped thousands of drug side effects to multiple labels, integrated the base predictors according to the weighted scoring ensemble strategy, and finally obtained a high-precision ensemble model [18]. Raies et al. used binary relevance and classifier chains methods to predict multiple toxicity endpoints for the same compound, and the comparative calculation results showed that the classifier chain algorithm achieved better performance [19]. Su et al. developed a series of models for hepatotoxicity prediction, where dose information and biological context were sufficiently explored, and provided a fitting method of dose-response curve [20, 21]. Jinwoo et al. employed gene-expression data, explored co-occurrences of pathologies, and proposed an integrative model to predict multiple organ pathologies, which is an advanced method to predict multiple pathology to the best of our knowledge [22]. The integrative model built a KNN classifier for each pathology and extracted the pathology associations to calculate the final scores. The accuracy of the prediction model ranges from 80% to 97% in both liver and kidney.

After review of recent research, we conclude that despite high accuracy performances achieved by several studies, existing works still have limitations. Firstly, most of prior studies focused on the prediction of toxicity (toxic or non-toxic) in a certain organ. Pathological finding prediction has not been explored much considering its importance for toxicity assessment. The handful existing pathology predictive models developed individual model for each label [19], which neglected the fact that compounds might cause several toxic effects simultaneously. Secondly, the pathological finding prediction is a multi-label classification task. In recent years, many multi-label classification methods have been proposed in the field of biological information [23, 24]. Nevertheless, most of the existing multi-label classification models in toxicogenomics area still used traditional machine learning models such as binary relevancy (BR) or classier chain (CC). The advanced deep learning technology has not been tested and employed. Directly applying existing deep learning network often obtains unsatisfactory results due to different characteristics of the toxicogenomics data, so it is required to build proper deep architecture with careful design. Lastly, some studies show very limited applicability for toxicity identification due to the adoption of small-scale, single-dose and single-time point data, thus data considering various factors should be fully adopted.

In this paper, we proposed a novel multi-label learning model, named Att-RethinkNet, for predicting drug-induced toxicity in multi-organ based on toxicogenomics data. Instead of handling the binary classification problem that differentiating whether a compound is toxic or non-toxic, we identified the specific pathological findings of liver and kidney, which is a multi-label learning task. To overcome the shortcomings such as ignoring label correlation in the traditional multi-label classification, inspired by Yang et al.'s work [25] which was evaluated on dozen multi-label data sets and justifies that RethinkNet obtains a better performance than state-of-the-art algorithms for multi-label classification tasks, we designed the deep framework

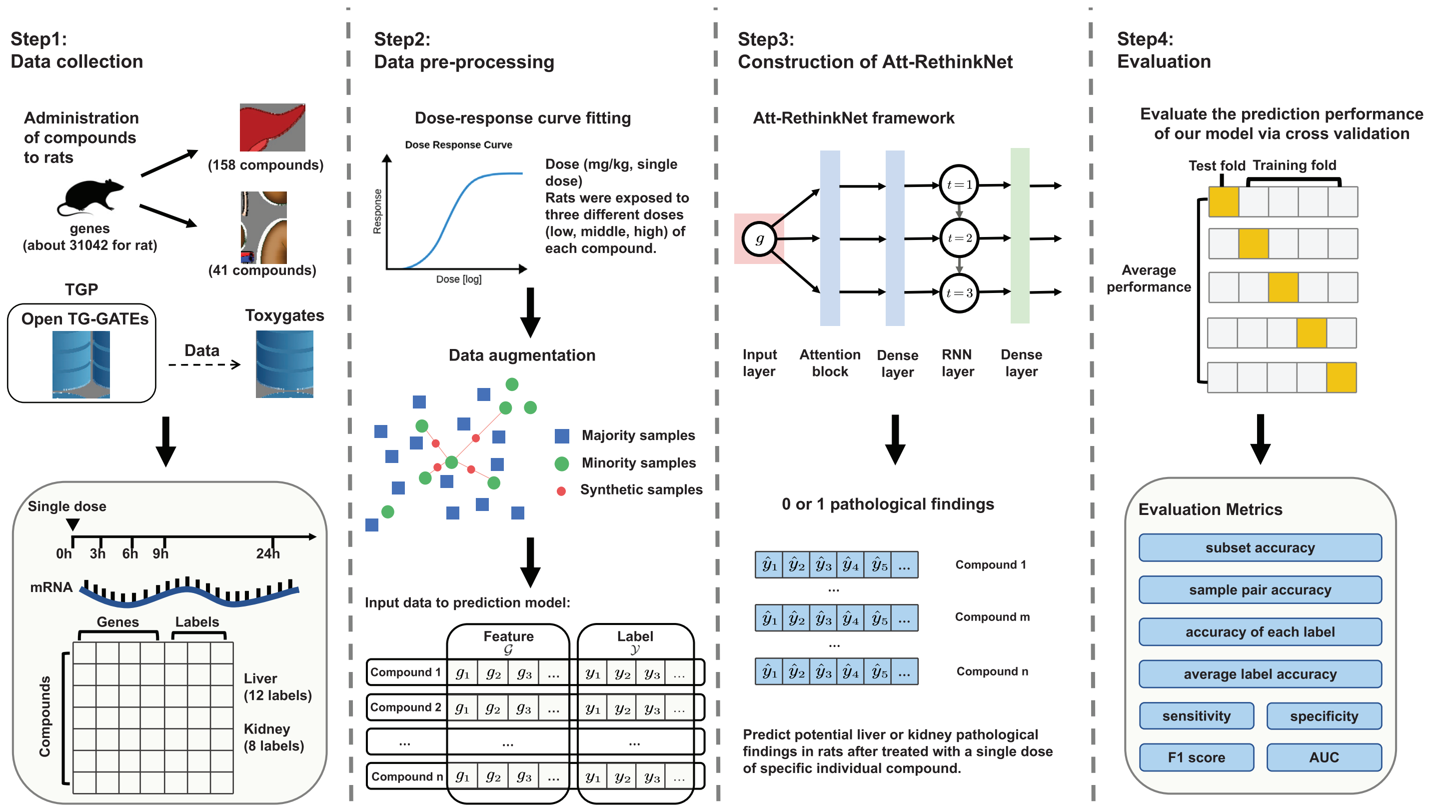

图 1. 提议方法的工作流程 方法。我们为所提出的方法设定了四个步骤：数据收集、数据预处理、所提出的A的构建 tt-RethinkNet 模型及 A 的评估 tt-RethinkNet。

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Att-RethinkNet 配备了记忆结构，可以有效利用标签相关性信息。此外，嵌入了注意力机制以关注重要特征并获得更好的特征表达。Att-RethinkNet 可应用于多个器官，并考虑复合类型、剂量和给药时间，比现有模型更为全面。更重要的是，它可以同时预测多种病理发现，而不是像以前的模型那样分别预测每种病理。根据我们所知，我们的 Att-RethinkNet 是首个基于深度架构探索多种病理发现的模型。在 Open TG-GATES 上的实验结果显示了该方法的有效性和高效性。

图1展示了本研究中预测病理学发现的概述。主要包括四个步骤：数据收集、数据预处理、提出的Att-RethinkNet模型的构建以及Att-RethinkNet的评估。每个步骤的详细内容将在以下材料与方法部分介绍。Att-RethinkNet的实现可以在 https://github.com/RanSuLab/Drug-Toxicity-Prediction-MultiLabel 获取。

材料与方法

第1步和第2步：数据收集和预处理

我们使用 Open TG-GATEs 来训练和验证我们的模型。TG-GATEs 是一个大规模的毒理学由日本毒理基因组学项目（TGP）开发的icogenomics数据库 [14]。

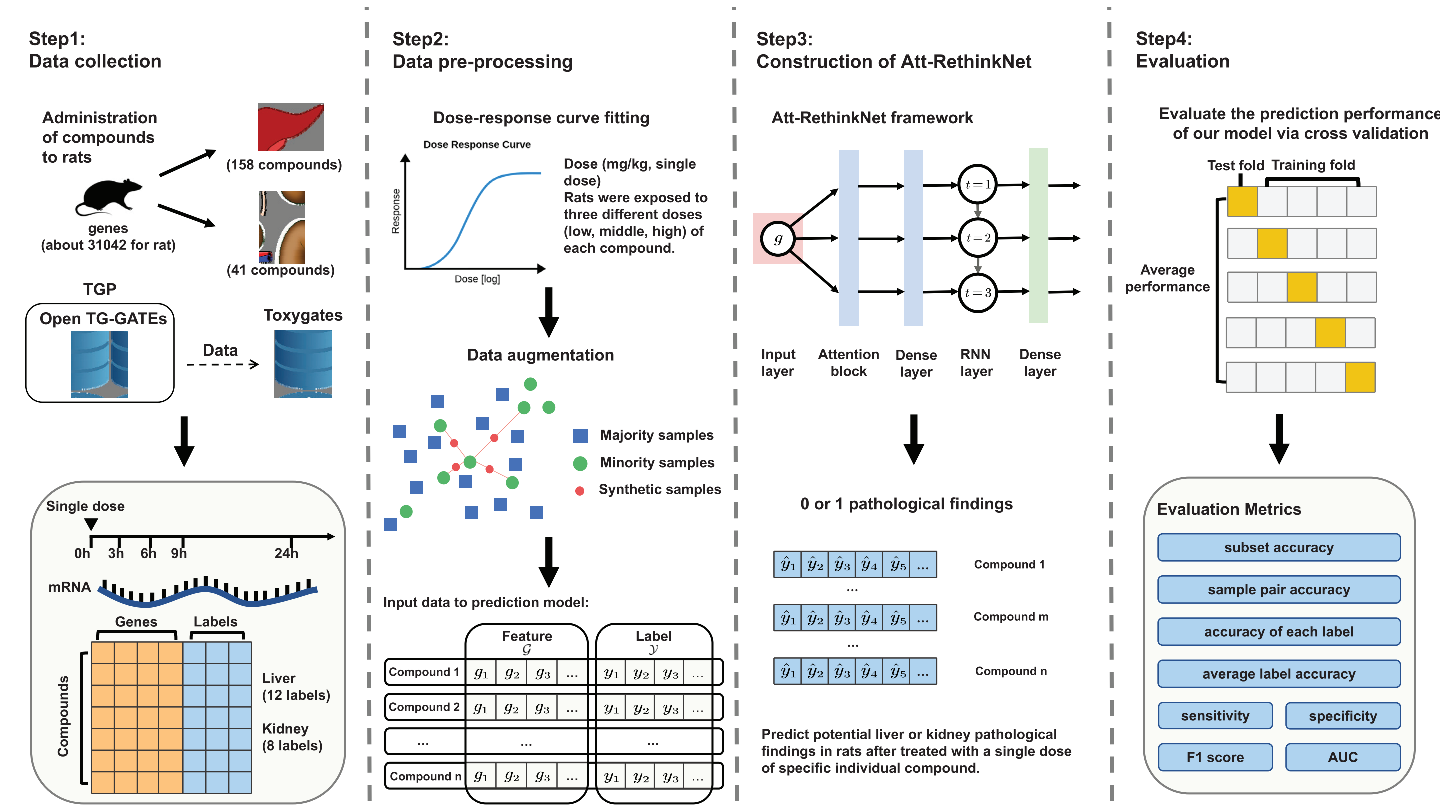

Fig 1. The work flow of the proposed method. We have four steps for the proposed method, data collection, data pre-processing, construction of the proposed Att-RethinkNet model and evaluation of the Att-RethinkNet.

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Att-RethinkNet, which is equipped with a memory structure and can effectively utilize the label correlation information. Besides, attention mechanism is embedded to focus on the important features and obtain better feature presentation. The Att-RethinkNet, which is applicable in multiple organs and takes account the compound type, dose, and administration time, is considerably more comprehensive than existing models. And more importantly, it predicts multiple pathological findings simultaneously, instead of predicting each pathology separately as the previous model did. To the best of our knowledge, our Att-RethinkNet is the first to explore the multiple pathological findings based on deep architecture. Experiment results on Open TG-GATES show the efficacy and efficiency of the proposed method.

Fig 1 shows an overview of predicting pathological findings in this study. There are mainly four steps, including data collection, data pre-processing, construction of the proposed Att-RethinkNet model and evaluation of the Att-RethinkNet. The details of each step will be introduced in the following materials and methods section. The implementation of the Att-RethinkNet can be found at https://github.com/RanSuLab/Drug-Toxicity-Prediction-MultiLabel.

Materials and methods

Step 1 & Step 2: Data collection and pre-processing

We used the Open TG-GATEs to train and validate our model. TG-GATEs is a large-scale toxicogenomics database developed by the Japanese Toxicogenomics Project (TGP) [14]. The

数据库包含170种化合物的基因表达谱和毒理学数据，这些数据来源于使用人类原代肝细胞和大鼠原代肝细胞的体外实验以及不同剂量和时间点的大鼠体内实验 [26, 27]。我们还使用了从Toxygates提取的数据，该平台作为一个集成、易于访问且用户友好的平台发布，用于Open TG-GATEs毒理基因组数据分析 [28]。Toxygates使用R中的Bioconductor affy软件包对每个样本的数据进行标准化。Toxygates使用户能够直接从Open TG-GATEs的原始微阵列数据中提取基因表达与变量（如剂量水平和暴露时间）之间的相关性，以人类可读的形式显示基因表达数据，并将CEL格式的二进制文件转换为CSV文件。

在我们的研究中，我们使用体内基因表达谱分析，对大鼠在所有三个剂量水平（低、中、高）下24小时的肝脏和肾脏进行了研究。对于肝脏数据，大鼠暴露于158种化合物，并收集了31,042个mRNA的表达水平。对于肾脏数据，大鼠暴露于41种化合物，也收集了31,042个mRNA的表达水平。肾脏数据的41种化合物全部包含在肝脏数据中，因此它们在两个器官上都进行了测试。然而，对于其他在肝脏中测试的化合物，其对肾脏的潜在病理风险尚不清楚。实验数据中涉及的药物或化学化合物显示在S1表中。接下来，我们从TG-GATEs中检查了四个时间点（3小时、6小时、9小时、24小时）体内的肝脏和肾脏病理变化，并只关注由大于或等于5种化合物引起的病理发现。

根据TG-GATEs，病理学家使用受控词汇描述了从体内实验中获得的药物诱导的病理症状。在我们的实验中，我们针对20个病理学发现，其中包括12个肝脏病理学发现，包括细胞浸润（CI）、嗜酸性变化（EC）、肥大（HY）、有丝分裂增加（IM）、未特定病变（NL）、微肉芽肿（MI）、坏死（NE）、肝膈结节（HN）、库普弗细胞增生（KCP）、单细胞坏死（SCN）、肿胀（SW）和细胞质空泡化（CV）；以及8个肾脏病理学发现，包括透明管型（HC）、淋巴细胞浸润（LCI）、嗜碱性变化（BC）、囊肿（CY）、扩张（DI）、囊性扩张（CD）、坏死（NE）和再生（RE）。对于多标签问题，标签向量由20个“1”或“0”组成，其中“1”表示存在病理学发现，“0”表示不存在病理学发现。

我们拟合了平滑的Sigmoid剂量-反应曲线，并从曲线中提取了最大反应值（R_{max}），该值包含全面的生物学信息，并在我们之前的研究中被证明是曲线的恰当表示 [20]。如果在三种剂量下基因表达值无法形成剂量-反应曲线，我们就会将这些基因移除，最终选择了6009个和8485个基因用于肝脏和肾脏。然后，根据我们收集的数据特征，我们改进了MLSMOTE（多标签合成少数类过采样技术）算法 [29]，以有效处理多标签分类的不平衡数据集，该方法可以克服多数类样本信息丢失的问题，并避免少数类样本复制导致的过拟合。在原方法中，我们增加了一个判断，以避免生成所有标签均为0的样本的罕见情况，从而在更大程度上丰富样本信息。步骤、计算公式a

database includes gene expression profiles and toxicological data of 170 compounds, derived from *in vitro* experiments using human primary hepatocytes and rat primary hepatocytes and *in vivo* experiments in rat at different dosages and time points [26, 27]. We also used data extracted from Toxygates which was released as an integrated, easily accessible and user-friendly platform for the Open TG-GATEs toxicogenomics data analysis [28]. Toxygates uses the Bioconductor *affy* package in R to carry out data normalization of each sample. Toxygates enables users to directly extract the correlation between gene expression and variables (such as dose level and exposure time) from the original microarray data of Open TG-GATEs, displays the gene expression data in human readable form, and convert the binary file in CEL format into CSV files.

In our studies, we used *in vivo* gene expression profiling of liver and kidney from rats at 24h of all three dose levels (low, middle, and high). For the liver data, rats were exposed to 158 compounds and expression levels of 31,042 mRNAs were collected. For the kidney data, rats were exposed to 41 compounds and also expression levels of 31,042 mRNAs were collected. The 41 compounds of kidney data were all included in the liver data so they were tested on both organs. However, for other compounds tested for liver, the potential pathological risk to kidney is unknown. The drugs or chemical compounds involved in the experimental data are shown in S1 Table. We next examined the *in vivo* hepatic and renal pathology taking place at four time points (3h, 6h, 9h, 24h) from TG-GATEs and focused only on the pathological findings that can be induced by larger than or equal to 5 compounds.

According to TG-GATEs, pathologists described drug-induced pathological symptoms obtained from *in vivo* tests using a controlled vocabulary. In our experiments, we targeted 20 pathological findings that comprise 12 liver pathological findings, including Cellular infiltration (CI), Eosinophilic change (EC), Hypertrophy (HY), Increased mitosis (IM), NOS lesion (NL), Microgranuloma (MI), Necrosis (NE), Hepatodiaphragmatic nodule (HN), Kupffer cell proliferation (KCP), Single cell necrosis (SCN), Swelling (SW), and Cytoplasmic vacuolization (CV) and 8 kidney pathological findings, including Hyaline cast (HC), Lymphocyte cellular infiltration (LCI), Basophilic change (BC), Cyst (CY), Dilatation (DI), Cystic dilatation (CD), Necrosis (NE), and Regeneration (RE). For multi-label problems, The label vector consists of 20 “1” or “0”, where the “1” represents a pathological finding exists, and “0” shows that the pathological findings does not exist.

We fitted a smooth sigmoid dose-response curve and extracted the maximum response (R_{max}) from the curve, which contained comprehensive biological information and was proved a proper presentation of the curve in our previous study [20]. We removed the genes if the expression values at three doses could not form the dose-response curve and finally 6009 and 8485 genes were picked for liver and kidney. Then, according to the characteristics of the data we collected, we improved the MLSMOTE (Multilabel Synthetic Minority Over-sampling Technique) algorithm [29] to effectively handle imbalanced data set for multi label classification, which can overcome the issue of information loss in majority class samples, and avoid over-fitting caused by replication of the minority class samples. We added a judgment in the original method to avoid the rare case of generating samples with all labels being 0, which can enrich the information of samples to a greater extent. The steps, calculation formulas and advantages of improved MLSMOTE algorithm was summarized in S1 Text. We also presented the results of metrics that measure the imbalance ratio of data set before and after the MLSMOTE in S1 Text, and indicated the number of samples liver or kidney had in the majority and minority classes of original data. After data augmentation, we obtained 16,460 samples and 16,268 samples for liver and kidney respectively.

步骤 3：Att-RethinkNet 的构建

传统多标签学习算法的综述。多标签学习旨在训练模型来解决每个样本同时关联多个标签的问题。在本节中，我们回顾了两种常用的多标签分类方法：二元相关（BR）和分类器链（CC）。BR和CC都将多标签分类任务分解为多个二分类问题。BR将每个标签的预测视为一个独立的二分类问题，其中每个分类器由所有特征训练，并且只需预测单个标签。由于每个标签都是单独处理的，该算法忽略了训练数据中标签之间可能存在的相关性。CC是BR的扩展。在CC方法中，根据标签顺序构建一系列二分类器，并将前面类别标签的二值分配作为附加特征处理。CC将标签加入特征空间

我们在研究中将所提出的方法与 BR 和 CC 算法进行了比较。对于每个二分类任务，我们选择逻辑回归（LR）、随机森林（RF）、线性支持向量机（SVM）作为基础分类器，并使用网格搜索策略优化 LR 的参数 C、RF 的两个超参数（树的数量和树的最大深度）、SVM 的参数 C。最后，我们通过五折交叉验证评估模型性能。

RethinkNet。RethinkNet 是一种用于多标签分类的深度学习架构，旨在模拟人类尝试探索标签之间相关性并通过反复思考同一问题直到能够消化的“再思考”过程，从而更有效地解决多标签问题。这个过程可以被视为一个序列预测问题。RethinkNet 的结构直观且易于理解。它由两层组成：循环神经网络（RNN）层和全连接（Dense）层。

RNN 层用于一个特定的目的：重新思考，这是一种迭代地优化预测结果的行为。RethinkNet 采用 RNN 来建模“重新思考”的过程，并充分利用 RNN 的记忆结构，该结构存储来自所有分类器的标签临时预测结果。所有分类器接收相同的信息，从而避免标签顺序的影响。不同于形成二元分类器链的 CC，RethinkNet 建立了多标签分类器链作为重新思考的序列。在全连接层中，层中的每个神经元都接收来自前一层（RNN 层）所有神经元的输入，并将前一 RNN 层的输出转化为所需的标签向量，从而生成最终的预测结果。RethinkNet 能够在最终预测前很好地考虑标签相关性。此外，该框架在学习阶段利用代价敏感的重加权损失函数，并在损失函数中对每个标签进行加权。

所提出的方法：Att-RethinkNet。我们提出的 Att-RethinkNet 是一种用于多标签分类的新型深度学习架构，其设计基于 RethinkNet。为了强调更重要的基因，我们在模型中嵌入了注意力机制。注意力机制的核心思想是从现有数据中学习权重分布，然后关注更重要的特征，这使得网络能够获得更好的特征表示。

在我们的实验中，我们在输入层和RNN层之间实现了注意力机制。为了提高性能，我们在所提出的Att-RethinkNet框架中使用了RNN的改进版本——长短期记忆（LSTM）网络。我们提出的Att-RethinkNet的架构包括输入层、注意力模块、RNN层和全连接层，如图2(a)所示。目标是从给定数据中学习一个函数 f： →

Step 3: Construction of Att-RethinkNet

Review of traditional multi-label learning algorithms. Multi-label learning aims at training models to tackle problems where each sample is associated with multiple labels simultaneously. We here reviewed two commonly used multi-label classification methods, binary relevance (BR) and classifier chains (CC) in this section. Both BR and CC decompose the multi-label classification task into multiple binary classification problems. BR treats the prediction of each label as an independent binary classification problem, where each classifier is trained by all the features and only a single label needs to be predicted. Since each label is treated individually, this algorithm ignores possible correlations among the labels of the training data. CC is an extension of BR. In the CC approach, a series of binary classifiers are constructed according to label order and the binary assignments of preceding class labels are treated as the additional features [30]. CC adds labels into feature space, so the relationship among classified labels can be considered in the rest classifiers, which overcomes the weakness of BR and usually reports a better performance.

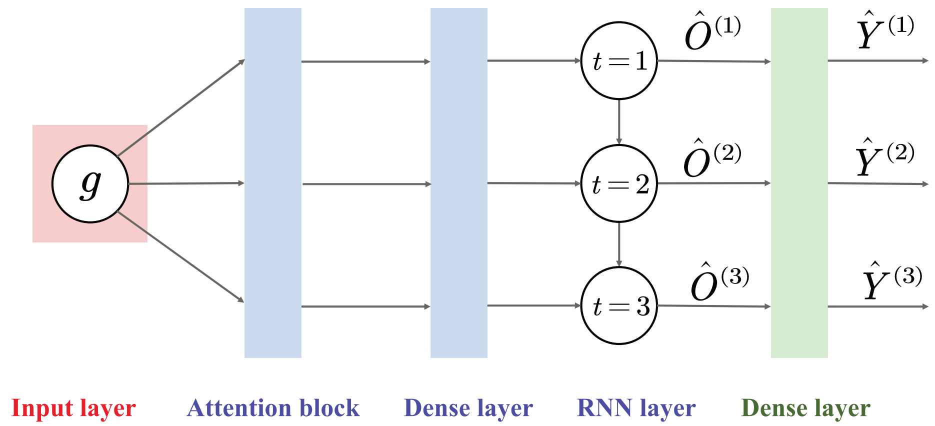
We compared the proposed method with BR and CC algorithms in our studies. For every binary classification task, we choose logistic regression (LR), random forest (RF), linear support vector machines (SVM) as base classifiers, and optimized the parameter C of LR, two hyper parameters *number of trees* and *maximum depth of trees* of RF, the parameter C of SVM using grid searching strategy. Finally, we evaluated the model performance via five-fold cross-validation.

RethinkNet. RethinkNet, a deep learning architecture for multi-label classification, is designed to mimic the "rethinking" process that human beings attempt to explore correlation between labels and solve multi-label problems more effectively through thinking the same issue over and over again until it is digestible. This process can be taken as a sequence prediction problem. The structure of RethinkNet is intuitive and understandable. It consists of two layers: recurrent neural network (RNN) layer and dense (fully connected) layer.

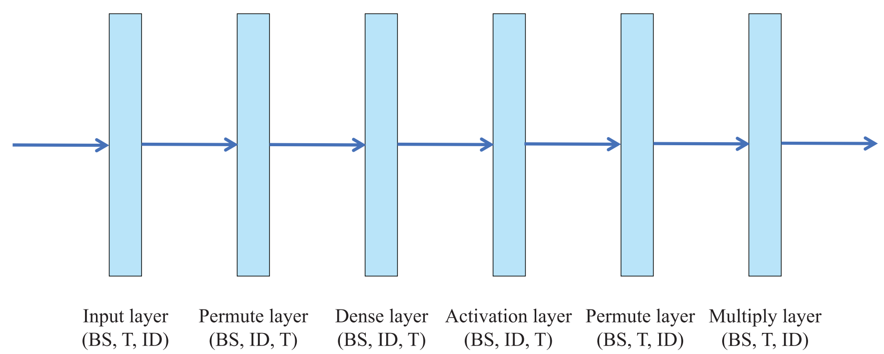
The RNN layer is used for a specific purpose: rethinking, an action that polishes the prediction result iteratively. RethinkNet adopts RNN to model the "rethinking" process and fully utilizes the RNN memory structure which stores temporary predictions on the labels from all classifiers. All classifiers receive the same information avoiding influence of the label order. Different from the CC which forms a chain of binary classifiers, a chain of multi-label classifiers as a sequence of rethinking is established [31]. On the dense layer, each neuron in the layer receives input from all the neurons present in its previous layer (the RNN layer) and transforms the output of previous RNN layer into the desired label vector, which generates the final prediction results. RethinkNet can well consider label correlation before the final prediction. Besides, the framework leverages cost-sensitive re-weighted loss function during learning phase and weights each label in the loss function according to the importance of the label.

The proposed method: Att-RethinkNet. Our proposed Att-RethinkNet, a novel deep learning architecture for multi-label classification, was designed based on the RethinkNet. To emphasize the more important genes, we embedded the attention mechanism in the model. The core idea of attention mechanism is to learn a weight distribution from existing data and then focus on the more important features, which enables the network to obtain better feature representation.

In our experiment, we implemented the attention mechanism between input layer and the RNN layer. To improve the performance, we used a modified version of RNN, Long Short-Term Memory (LSTM) networks, in the proposed Att-RethinkNet framework. The architecture of our proposed Att-RethinkNet includes the input layer, attention block, RNN layer, and dense layer as shown in Fig 2(a). The goal is to learn a function $f : \mathcal{X} \rightarrow \mathcal{Y}$ from a given



(a) The architecture of the Att-RethinkNet framework.



(b) The structure of the attention block.

图2。（a）是Att-RethinkNet框架的架构，这是对图1中第3步所提出的深度神经网络结构的具体描述。（b）是Att-RethinkNet框架中注意力模块的结构。BS表示批量大小。T表示时间步，描绘每个LSTM单元的迭代次数，不影响参数数量。ID表示输入维度。第一个permute层重新组织输入层，交换输入的T和ID维度。dense层和激活层计算输入的注意力概率（权重），即计算每个基因特征对应的权重。激活层对激活的神经元应用softmax函数。第二个permute层也重新组织已计算权重的数据的维度，以便在multiply层中进行乘法操作。multiply层是注意力模块的最后一层。在此层中，注意力模块的输入

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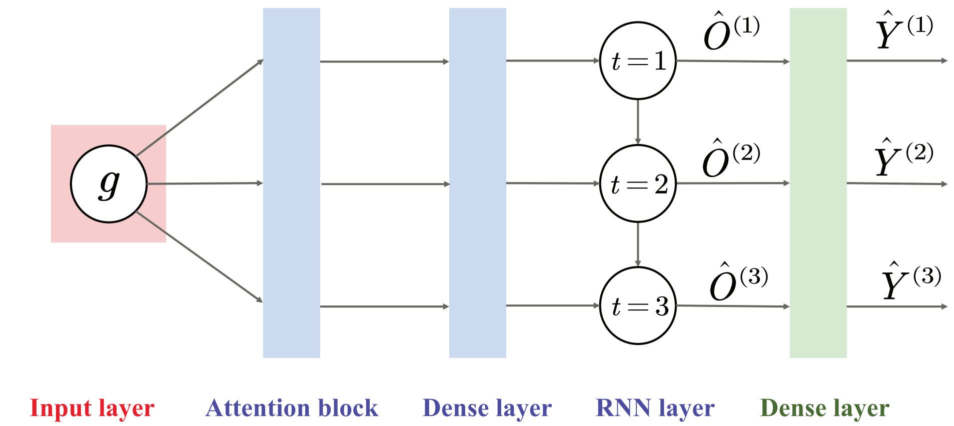
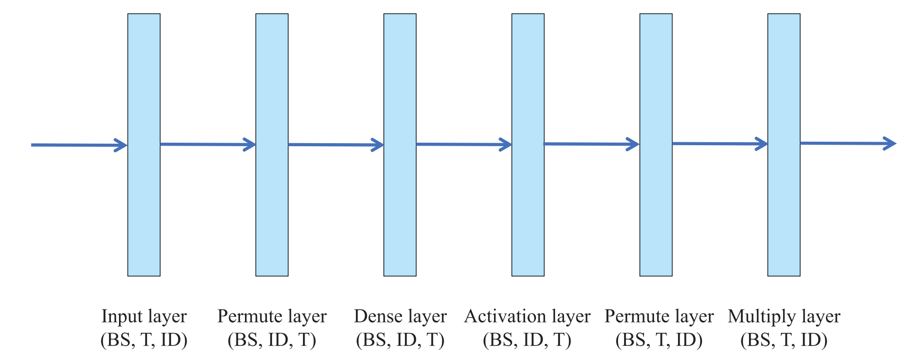
训练数据集并实现从特征向量 x ∈ � $\mathbb{R}^{dim}$ 到对应的病理结果标签 Y ∈ � f0, 1g^L 的映射。

输入层包含基因特征。RNN 层学习 T 次迭代，每次迭代表示一个思考过程。RNN 层在第 t 次迭代的输出简写为 ˆO$^{(t)}$，表示第 t 个预测标签向量 ˆY$^{(t)}$ 的嵌入。同样地，ˆO$^{(t)}$ 的信息将传递到 RNN 层的第 (t + 1) 次迭代，即 Att-RethinkNet 将使用前一次迭代的临时预测结果来获得更好的标签预测 ˆY$^{(t+1)}$。当执行完 T 次迭代后，ˆ Y$^{(T)}$ 是最终预测。ˆY$^{(T)}$ 是经过迭代修正后的准确标签集，这意味着难以预测的标签也有更高的概率被归入正确类别。在我们的实验中，我们为肝脏数据设置 T = 5，为肾脏数据设置 T = 3，原因是



(a) The architecture of the Att-RethinkNet framework.



(b) The structure of the attention block.

Fig 2. (a) is the architecture of the Att-RethinkNet framework, which is the specific description of our proposed deep neural network structure of Step3 in Fig 1. (b) is the structure of the attention block in Att-RethinkNet framework. BS means batch size. T means time step, depicting the number of iterations of each LSTM unit, without affecting the number of parameters. ID means input dimension. The first permute layer re-organizes the input layer, which permutes the T and ID dimensions of the input. The dense layer and the activation layer compute attention probabilities (the weight) for the input, that is, calculate the weight corresponding to each gene feature. The activation layer applies softmax function to the activated neurons. The second permute layer also re-organize the dimensions of the data whose weights have been calculated so that the multiplication can be operated in the multiply layer. The multiply layer is the last layer in attention block. In this layer, input of attention block times the probability vector of attention, achieving the weight allocation of feature vector. The attention mechanism assigns different importance to features which improves the result of classification greatly.

https://doi.org/10.1371/journal.pcbi.1010402.g002

training data set and realize the mapping from the feature vector $x \in \mathcal{X} \subseteq \mathbb{R}^{dim}$ to corresponding pathological findings label $Y \in \mathcal{Y} \subseteq \{0, 1\}^L$.

The input layer contains the gene feature. The RNN layer learns T iterations, and each iteration represents a thinking process. The output of RNN layer at t-th iteration is abbreviated as $\hat{O}^{(t)}$, which stands for the embedding of t-th prediction label vector $\hat{Y}^{(t)}$. By the same token, the information of $\hat{O}^{(t)}$ will be passed to $(t + 1)$-th iteration in the RNN layer, that is, Att-RethinkNet will use the temporary prediction results of the previous iteration to obtain better label predictions $\hat{Y}^{(t+1)}$. When T iterations are executed, $\hat{Y}^{(T)}$ is the final prediction. $\hat{Y}^{(T)}$ is an accurate set of labels that has been iteratively revised, which means labels that are difficult to predict will also have a greater probability of being classified into the correct category. In our experiments, we set $T = 5$ for liver data and $T = 3$ for kidney data, for the reason that the

我们提出的模型的性能通常在固定的第 T 次重新思考迭代时收敛。随着 T 的增加，病理发现的预测准确性基本没有变化。我们还在图 2(b) 中展示了注意力模块的结构。

用于预测药物引起的病理发现的拟议方法的伪代码 n
多器官样本如算法1所示。

算法 1 使用 K 折交叉验证预测药物诱导的病理学发现。输入：

1: 输入：涉及所有化合物和所有基因的基因表达数据。
输出：2：输出

t: 基于基因表达数据预测药物诱导病理发现的模型。3: 根据三个剂量水平（低、中、高）拟合剂量反应曲线，并选择合适的指标来表示曲线的完整生物学信息。4: 扩增和平衡数据。5: 数据归一化。6: 对 i = 1; i < K; i ++ 执行循环 7: 将扩增后的数据集 D 划分为测试集 D_{test} 和训练集 D_{train}。8: 将 D_{train} 输入到 Att-RethinkNet 框架。9: 计算注意力概率并为所有基因加权。10: 预测多个器官的潜在病理发现，并迭代修改临时预测结果。11: 在 D $_{test}$ 上测试并记录结果。12: 循环结束

13: 计算 K 个结果的平均值，获得最终评估结果，并分析我们提出模型的分类效果。

步骤4：评估

在本文中，所有实验均通过五折交叉验证进行评估。在单标签分类中，可以使用传统的评估指标。在多标签分类中，一个样本可能只有部分标签被正确分类，因此评估指标需要能够客观地反映多标签分类器的性能 [32–34]。因此，我们使用了两组评估指标，一类是基于样本的指标，它们分别计算每个样本的性能，然后对所有样本取平均值；另一类是基于标签的指标，它们从每个标签的角度进行评估，然后对所有标签取宏平均或微平均 [35]。基于样本的指标包括子集准确率 (ACC)、样本对准确率 (ACC_{pair}) 以及每个标签的准确率 (ACC_{lab})；基于标签的指标包括平均标签准确率 (ACC_{avelab})、宏敏感性 (SEN)、宏特异性 (SPE) 以及宏 F1 分数 (F1)，均用于评估我们的模型。假设 x 是一个样本

$$\mathrm{ACC} = \frac{1}{n}\sum_{i=1}^{n}\Phi(Y_i = \hat{Y}(x_i)) \tag{1}$$

哪里

$$\Phi(\cdot) = \begin{cases} 1, & \cdot \text{ is true (if and only if } \hat{Y} \text{ exactly matches } \hat{Y}(x_i)), \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

performance of our proposed model generally converges at the fixed *T*th iteration of rethinking. With the increase of *T*, the prediction accuracy of pathological findings basically did not change. We also show the structure of the attention block in Fig 2(b).

The pseudo-code of the proposed method to predict drug-induced pathological findings in multi-organ samples is shown in Algorithm 1.

Algorithm 1 Drug-induced pathological finding prediction using K-fold cross validation.

```
Input:
1: Input: Gene expression data involving all compounds and all genes.
Output:
2: Output: Model to predict drug-induced pathological findings based
on gene expression data.
3: Fit the dose-response curve based on three dose levels (low, middle
and high), and select a proper measure to represent the full biologi-
cal information of the curve.
4: Augment and balance the data.
5: Data normalization.
6: for i = 1; i < K; i++ do
7:   Divide the augmented data set D into test set D_test and training
set D_train.
8:   Feed D_train into Att-RethinkNet framework.
9:   Calculate attention probabilities and weight all genes
10:   Predict the potential pathological findings in multiple organs,
and modify temporary prediction results iteratively.
11:   Test on D_test and record the results.
12: end for
13: Calculate the average of the K results, obtain the final evalua-
tion results, and analyze the classification effect of our proposed
model.
```

Step 4: Evaluation

In this paper, all experiments were evaluated by five-fold cross-validation. In single-label classification, the traditional evaluation metrics can be used. In multi-label classification, a sample may have part of labels classified correctly, so evaluation measures are required to have an objective view of the performance of multi-label classifiers [32–34]. Therefore, we used two groups of evaluation metrics, one is sample-based metrics that compute the performance of each sample separately and then average it over all samples and the other is label-based metrics that conduct the evaluation in terms of each label and then take the macro/micro average over all labels [35]. Sample-based metrics including subset accuracy (ACC), sample pair accuracy (ACC_{pair}) and accuracy of each label (ACC_{lab}) and label-based metrics including average label accuracy (ACC_{avelab}), macro sensitivity (SEN), macro specificity (SPE) and macro F1 score (F1) were used to evaluate our model. Assuming *x* is a sample, *n* is the number of test sample, and Y_i and $\hat{Y}(x_i)$ represent the true and predicted label vector for the *i*th sample, respectively, these metrics are defined as follows:

$$\mathrm{ACC} = \frac{1}{n}\sum_{i=1}^{n}\Phi(Y_i = \hat{Y}(x_i)) \tag{1}$$

Where

$$\Phi(\cdot) = \begin{cases} 1, & \cdot \text{ is true (if and only if } \hat{Y} \text{ exactly matches } \hat{Y}(x_i)), \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

$$\mathrm{ACC_{pair}} = \frac{1}{n}\sum_{i=1}^{n}\frac{|Y_i \cap \hat{Y}(x_i)|}{|Y_i \cup \hat{Y}(x_i)|} \quad (3)$$

$$\mathrm{ACC_{lab(l)}} = \frac{1}{n}\sum_{i=1}^{n}[[\hat{y}_i^l = y_i^l]] \quad (4)$$

子集准确率是指预测标签向量与真实标签向量相同的样本的比例。对于测试集的预测标签向量，分类结果只有在预测值与标签集的真实值完全相同时才被认为是正确的。$\mathrm{ACC^{pair}}$ 反映部分正确的程度，比子集准确率更宽松。$\mathrm{ACC_{lab}}$ 表示每个标签的准确率，通过它我们可以找出哪些病理发现容易识别 [33]。在计算基于标签的指标时，标签l的基本统计量真阳性（TP）、假阳性（FP）、真阴性（TN）和假阴性（FN）定义如下：

$$\mathrm{TP}_l = |\{x_i \mid y_l \in Y_i \wedge y_l \in \hat{Y}(x_i), 1 \le i \le n, 1 \le l \le L\}| \quad (5)$$

$$\mathrm{FP}_l = |\{x_i \mid y_l \notin Y_i \wedge y_l \in \hat{Y}(x_i), 1 \le i \le n, 1 \le l \le L\}| \quad (6)$$

$$\mathrm{TN}_l = |\{x_i \mid y_l \notin Y_i \wedge y_l \notin \hat{Y}(x_i), 1 \le i \le n, 1 \le l \le L\}| \quad (7)$$

$$\mathrm{FN}_l = |\{x_i \mid y_l \in Y_i \wedge y_l \notin \hat{Y}(x_i), 1 \le i \le n, 1 \le l \le L\}| \quad (8)$$

$$\mathrm{ACC_{avelab}} = \frac{1}{L}\sum_{l=1}^{L}\frac{\mathrm{TP}_l + \mathrm{TN}_l}{\mathrm{TP}_l + \mathrm{TN}_l + \mathrm{FP}_l + \mathrm{FN}_l} \quad (9)$$

$$\mathrm{SEN} = \frac{1}{L}\sum_{l=1}^{L}\frac{\mathrm{TP}_l}{\mathrm{TP}_l + \mathrm{FN}_l} \quad (10)$$

$$\mathrm{SPE} = \frac{1}{L}\sum_{l=1}^{L}\frac{\mathrm{TN}_l}{\mathrm{TN}_l + \mathrm{FP}_l} \quad (11)$$

$$\mathrm{F1} = \frac{1}{L}\sum_{l=1}^{L}\frac{2\mathrm{TP}_l}{2\mathrm{TP}_l + \mathrm{FP}_l + \mathrm{FN}_l} \quad (12)$$

这里 L 和 y_l 分别表示标签的数量和第 l 个真实标签。此外，我们还采用了接收器工作特性曲线（ROC）和曲线下面积（AUC）来获得对评估和测量的多重视角。

结果

首先，我们检查了基于 LSTM 的 Att-RethinkNet 所产生的特征以及结果混淆矩阵。然后，我们讨论了 RNN 层中 LSTM 和 SRN 算法的预测性能。我们将所提出的方法与原始 RethinkNet 以及传统的 BR 和 CC 进行了比较。此外，我们还与 Kim 等人提出的综合模型进行了比较，该模型是药物引起的病理发现预测的最先进工作。我们在 Open TG-GATES 体内肝脏数据上进行了所有实验，并

$$\mathrm{ACC_{pair}} = \frac{1}{n}\sum_{i=1}^{n}\frac{|Y_i \cap \hat{Y}(x_i)|}{|Y_i \cup \hat{Y}(x_i)|} \quad (3)$$

$$\mathrm{ACC_{lab(l)}} = \frac{1}{n}\sum_{i=1}^{n}[[\hat{y}_i^l = y_i^l]] \quad (4)$$

Subset accuracy is the fraction of samples whose predicted label vector is the same as the true label vector. For a predicted label vector of a test set, the classification result is considered to be correct if and only if the prediction value is exactly equal to the true value of label set. $\mathrm{ACC_{pair}}$ reflects the degree of partial correctness, which is more lenient than subset accuracy. $\mathrm{ACC_{lab}}$ represents the accuracy of each label, by which we can find which pathological finding is easy to identify [33]. When calculating label-based metrics, the basic statistics true positive (TP), false positive (FP), true negative (TN), and false negative (FN) for label *l* is defined as follow:

$$\mathrm{TP}_l = |\{x_i \mid y_l \in Y_i \wedge y_l \in \hat{Y}(x_i), 1 \le i \le n, 1 \le l \le L\}| \quad (5)$$

$$\mathrm{FP}_l = |\{x_i \mid y_l \notin Y_i \wedge y_l \in \hat{Y}(x_i), 1 \le i \le n, 1 \le l \le L\}| \quad (6)$$

$$\mathrm{TN}_l = |\{x_i \mid y_l \notin Y_i \wedge y_l \notin \hat{Y}(x_i), 1 \le i \le n, 1 \le l \le L\}| \quad (7)$$

$$\mathrm{FN}_l = |\{x_i \mid y_l \in Y_i \wedge y_l \notin \hat{Y}(x_i), 1 \le i \le n, 1 \le l \le L\}| \quad (8)$$

$$\mathrm{ACC_{avelab}} = \frac{1}{L}\sum_{l=1}^{L}\frac{\mathrm{TP}_l + \mathrm{TN}_l}{\mathrm{TP}_l + \mathrm{TN}_l + \mathrm{FP}_l + \mathrm{FN}_l} \quad (9)$$

$$\mathrm{SEN} = \frac{1}{L}\sum_{l=1}^{L}\frac{\mathrm{TP}_l}{\mathrm{TP}_l + \mathrm{FN}_l} \quad (10)$$

$$\mathrm{SPE} = \frac{1}{L}\sum_{l=1}^{L}\frac{\mathrm{TN}_l}{\mathrm{TN}_l + \mathrm{FP}_l} \quad (11)$$

$$\mathrm{F1} = \frac{1}{L}\sum_{l=1}^{L}\frac{2\mathrm{TP}_l}{2\mathrm{TP}_l + \mathrm{FP}_l + \mathrm{FN}_l} \quad (12)$$

Here L and y_l denote the number of labels and the lth true label, respectively. Additionally, we also adopted the receiver operating characteristics curve (ROC) and area under the curves (AUC) to get a multiple perspective on evaluation and assessment.

Results

Here we firstly look into the features produced by the Att-RethinkNet based on LSTM and the outcome confusion matrix. Then we discussed the prediction performance of LSTM and SRN algorithms of the RNN layer. We compared the proposed method with the original RethinkNet and the traditional BR and CC. Additionally, we compared with the integrative model proposed by Kim et al. [22], which is the state-of-the-art work for drug-induced pathological finding prediction. We conducted all the experiments on Open TG-GATES *in vivo* liver and

肾脏数据。此外，为了进一步验证模型的泛化能力，我们使用了一个在体外大鼠数据集上未见过的独立测试集。

预测结果的可视化和混淆矩阵

首先，我们进行了t分布随机邻域嵌入（t-SNE）以在低维空间中可视化数据。原始特征（基因）和RNN层之后生成的特征如图3所示。这里我们展示了肝脏数据中的病理学发现：细胞浸润、坏死和库普弗细胞增生，以及肾脏数据中的囊肿、淋巴细胞浸润和坏死。

从图3可以看出，使用原始特征的正负样本是混合的，并且重叠很多。但经过Att-RethinkNet后，0–1类别可以更好地区分。这表明生成的特征比原始特征更具区分性和信息量。

为了更细致地理解所提出模型的结果，我们在图4中展示了病理分类结果的混淆矩阵。行和列的数值分别表示测试数据的真实标签和预测标签。从混淆矩阵可以清楚地看出，与TP和TN相比，模型的FP和FN值相当小，因此FP和FN率很低。这显示了所提出模型的出色性能。

Att-RethinkNet 与 LSTM 和 SRN 的比较

在我们的实验中，Att-RethinkNet 的 RNN 层采用了 LSTM 网络。它的优势在于，它不仅将多个相关的病理发现附加到输入数据上，并通过记忆机制存储前期操作的临时预测，还可以通过遗忘门选择性地忘记先前标签的预测。为了证明应用 LSTM 算法改善分类效果的有效性，我们比较分析了在 RNN 层中使用 LSTM 和简单循环网络 (SRN) 的方法。

表1列出了两种算法在肝脏数据和肾脏数据上的分类结果。在预测肝毒性病理发现时，由LSTM算法实现的所提出的Att-RethinkNet模型获得了89.4%的ACC，比使用SRN的模型高出1.9%，并且除了SPE外，在所有评估指标上都取得了更高的数值。肾毒性病理发现的分类结果显示，LSTM的分类结果也高于SRN，ACC的提升水平超过1.0%。

分类准确率有希望的原因在于 LSTM 内的门结构以及内部复杂的训练参数提高了模型对长序列数据的处理能力，并避免了 RNN 中梯度消失的问题。更具体地说，在构建用于预测多器官药物诱导病理发现的 Att-RethinkNet 的过程中，LSTM 算法为重新思考 RNN 层提供了一种新的改进策略。当通过一次迭代获得一组病理发现预测标签时，其中一部分作为当前迭代的临时结果产生，另一部分信息继续传递。在下一次迭代开始时，LSTM 不再直接使用前一次迭代的结果进行更好的预测，而是通过遗忘门确定信息的遗忘程度。最终，通过选择性的 T 次迭代思考，我们基于 L 的 Att-RethinkNet 模型

kidney data. In addition, in order to further verify the generalization of the model, we used an unseen and independent test set on the *in vitro* data set of rats.

Visualization and confusion matrix of the prediction results

Firstly, we performed t-distributed stochastic neighbor embedding (t-SNE) to visualize the data in a low dimension space. Raw features (genes) and features produced after RNN layer are shown in Fig 3. Here we show pathological findings cellular infiltration, necrosis and kupffer cell proliferation from liver data and cyst, lymphocyte cellular infiltration and necrosis from kidney data.

As can be seen from Fig 3, positive and negative samples with the raw features are mixed and have much overlapping. But after the Att-RethinkNet, the 0–1 classes can be better separated. This has indicated that the generated features are more distinctive and informative than the raw features.

To have a more granular understanding of the results of the proposed model, we show the confusion matrix of the pathology classification results in Fig 4. The values of the rows and columns represent the true and predicted labels on test data, respectively. From the confusion matrix, it is clear that the model has quite small values of FP and FN compared to TP and TN, and therefore low FP and FN rate. This has shown an impressive performance of the proposed model.

Comparison between Att-RethinkNet with LSTM and SRN

In our experiment, the RNN layer of Att-RethinkNet adopts the LSTM network. Its advantage is that it not only attaches multiple relevant pathological findings to an input data and stores temporary predictions from earlier operations through memory mechanism, but also selectively forgets the prediction of previous labels through forget gate. In order to prove the effectiveness of applying LSTM algorithm in improving the classification effect, we compared and analyzed the methods of using LSTM and simple recurrent network (SRN) in RNN layer.
Table 1 lists the classification results of the two algorithms on the liver data and kidney data. In the prediction of pathological findings of hepatotoxicity, the proposed Att-RethinkNet model implemented by LSTM algorithm obtained an ACC of 89.4%, which was 1.9% higher than that of the model using SRN, and obtained higher values in all evaluation metrics except SPE. The classification results of nephrotoxic pathological findings showed that the classification results of LSTM were also higher than SRN, and the improvement level of ACC exceeded 1.0%.

The reason for the promising classification accuracy is that the gate structure within LSTM and the internal complex training parameters improve the processing ability of the model for long sequence data and avoid the problem of vanishing gradients in RNN. More specifically, in the process of building Att-RethinkNet for predicting drug-induced pathological findings in multiple organ, LSTM algorithm provides a new improvement strategy for rethinking of RNN layer. When a group of pathological finding prediction labels are obtained through one iteration, one part is produced as the temporary result of the current iteration, and the other part of the information continues to be transmitted. And at the beginning of the next iteration, LSTM no longer directly uses the results of the previous iteration for better prediction, but determines the forgetting degree of the information through the forget gate. Finally, through selective T times iterative thinking, our Att-RethinkNet model based on LSTM can better analyze the implicit association between gene expression data and corresponding pathological findings, as well as the internal impact between different pathological findings, and then iteratively polish the multi-label prediction results, provide a more accurate label set and show more accurate classification results.

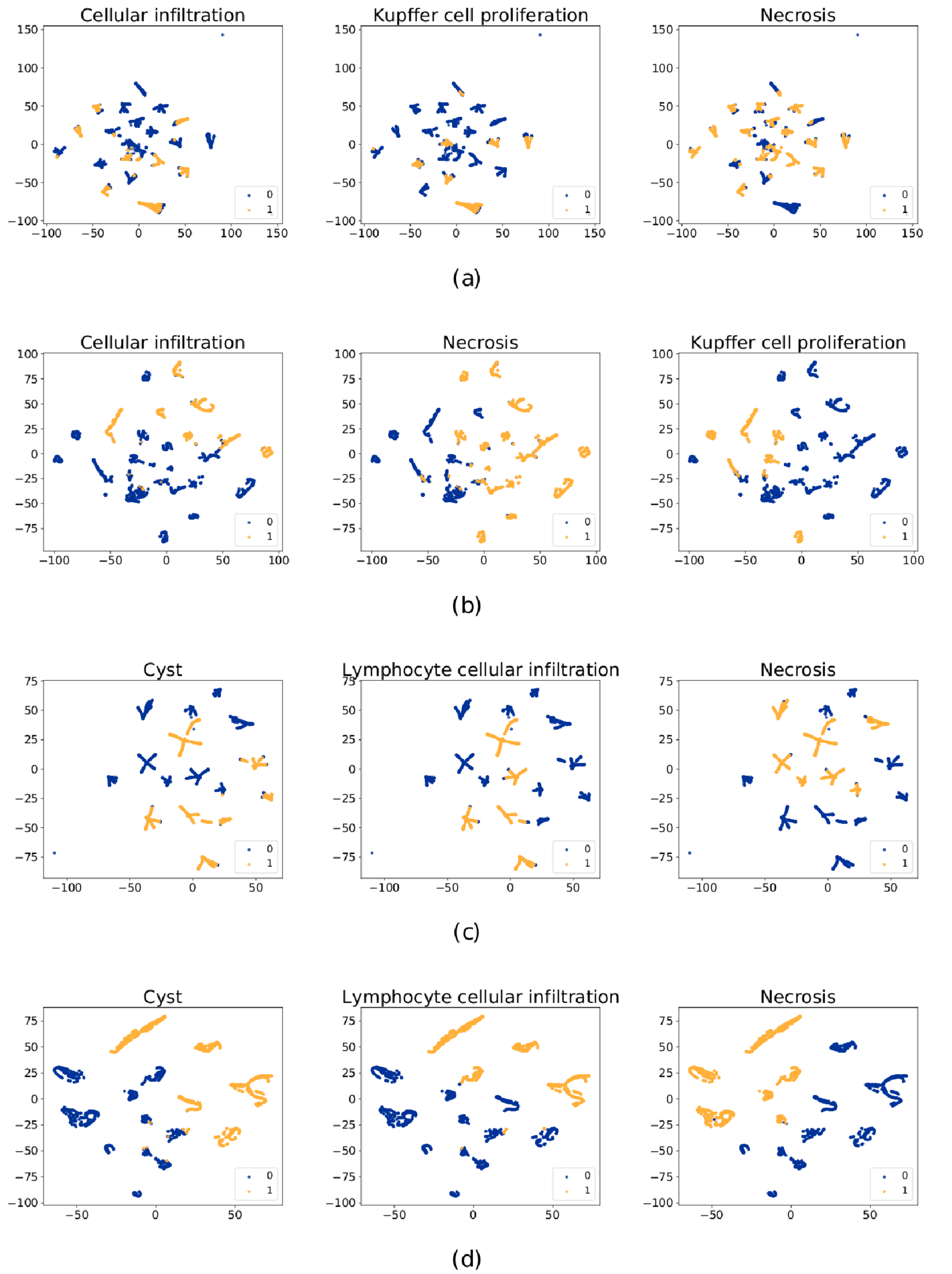


图3. 不同病理发现的特征t-SNE可视化（来自同一fold）。(a)和(b)分别显示原始特征和经过RNN层后生成的特征。涉及肝脏数据中的三种病理发现：细胞浸润、坏死和库普弗细胞增生。(c)和(d)分别显示原始特征和经过RNN层后生成的特征，涉及肾脏数据中的三种病理发现：囊肿、淋巴细胞浸润和坏死。蓝色点表示0（无发现），黄色点表示1（有发现）。使用t-SNE对所有目标病理发现的可视化可以在S1图（肝脏）和S2图（肾脏）中找到。

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表 2 和表 3 显示了我们提出的模型在肝脏和肾脏数据集上 RNN 层的 ACC_{lab} 值，分别使用两种算法。可以看出，基于 LSTM 的 Att-RethinkNet 在药物测试集上具有更高的 ACC_{lab}，20 个病理标签的预测准确率基本都超过 97%，这表明基于 LSTM 的 Att-RethinkNet

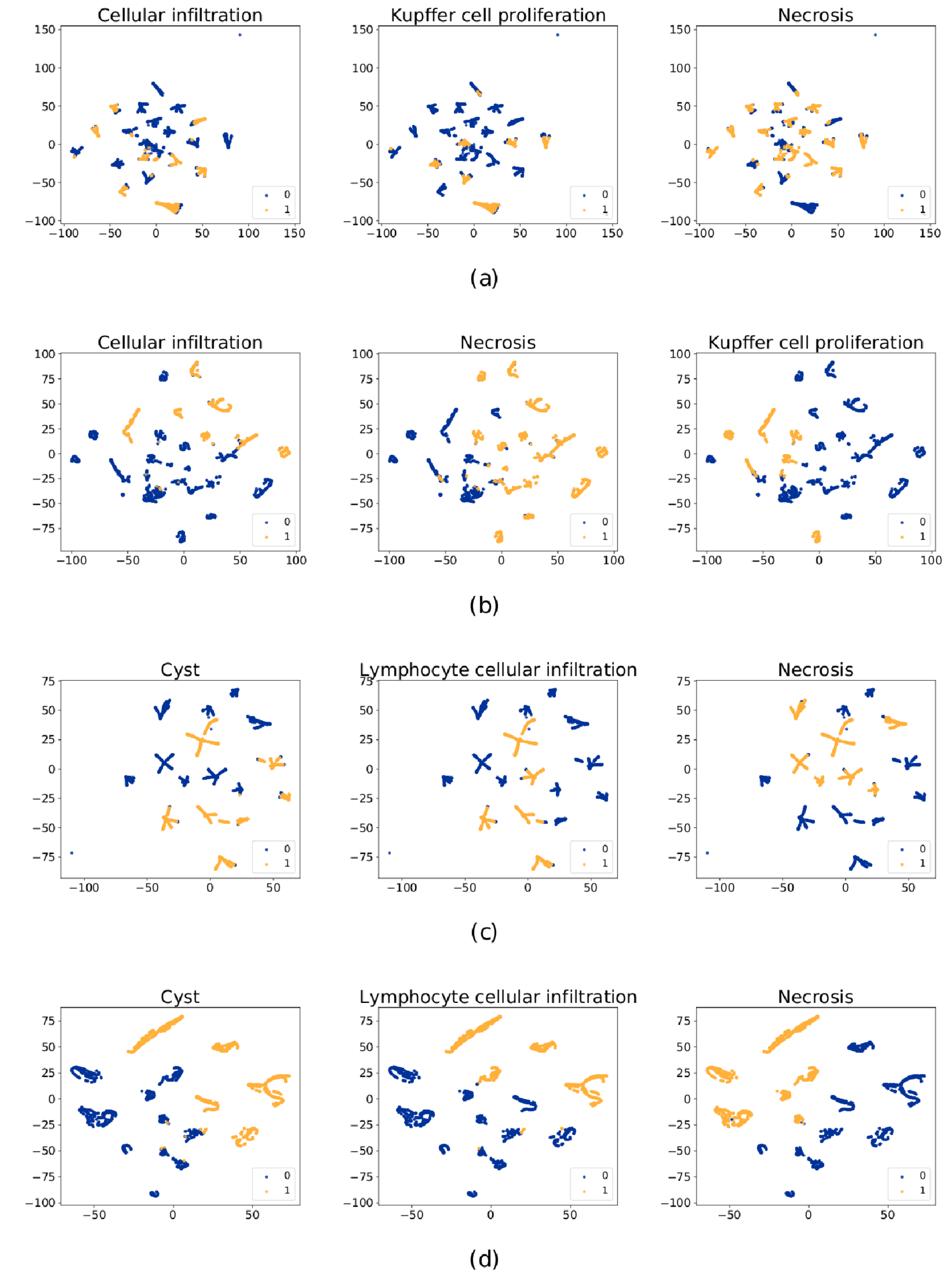
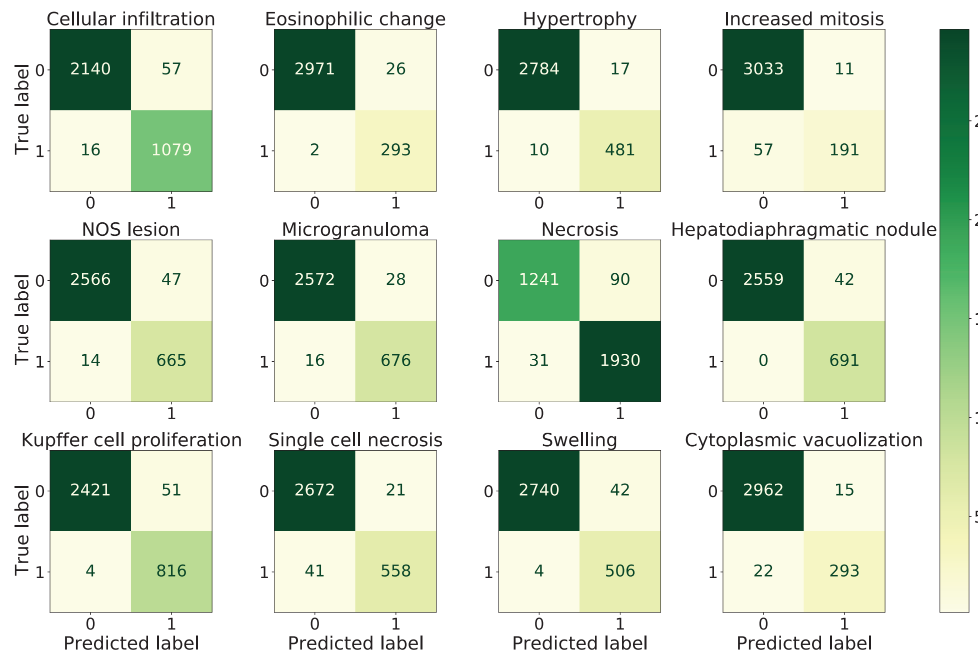


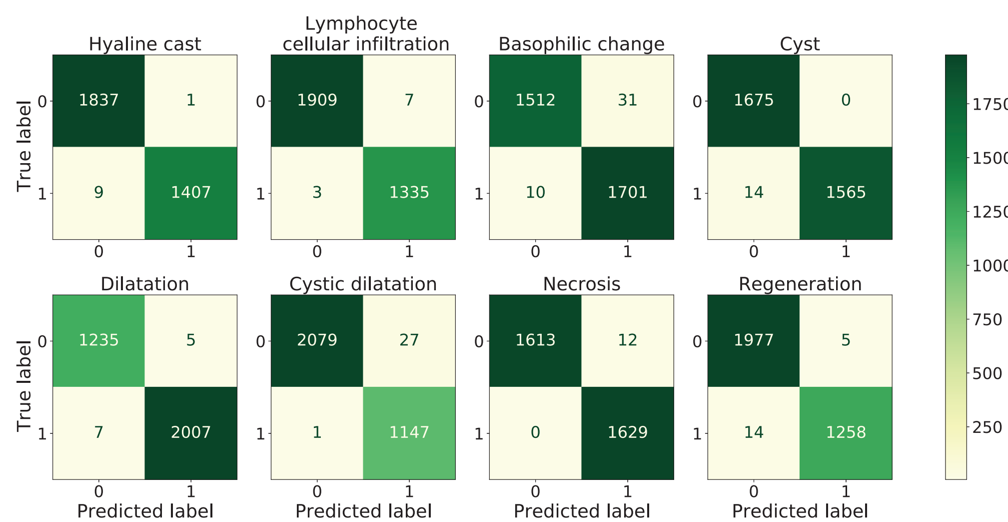
Fig 3. The t-SNE visualization of features for different pathological findings (from one fold). (a) and (b) shows the raw features and features generated after RNN layer, respectively. Three pathological findings cellular infiltration, necrosis and kupffer cell proliferation from liver data are involved. (c) and (d) shows the raw features and features generated after RNN layer, respectively and three pathological findings cyst, lymphocyte cellular infiltration and necrosis from kidney data are involved. The blue points represent 0 (no findings) and the yellow points represent 1 (with findings). The visualization of all targeted pathological findings using t-SNE can be found in S1 Fig for liver and S2 Fig for kidney.

https://doi.org/10.1371/journal.pcbi.1010402.g003

Tables 2 and 3 show the ACC_{lab} values of the RNN layer of our proposed model in the liver and kidney data sets using two algorithms respectively. It can be seen that Att-RethinkNet based on LSTM has higher ACC_{lab} for the drug test set, and the prediction accuracy of 20 pathological labels is basically more than 97%, which shows that Att-RethinkNet based on LSTM



(a) Liver



(b) Kidney

图4. 病理分类的混淆矩阵。左上角表示 TN，右上角表示 FP，左下角是 FN，右下角是 TP。(a) 显示肝脏数据的混淆矩阵，(b) 显示肾脏数据的混淆矩阵。结果来自第一折。其他折的结果显示在 S3 图中。

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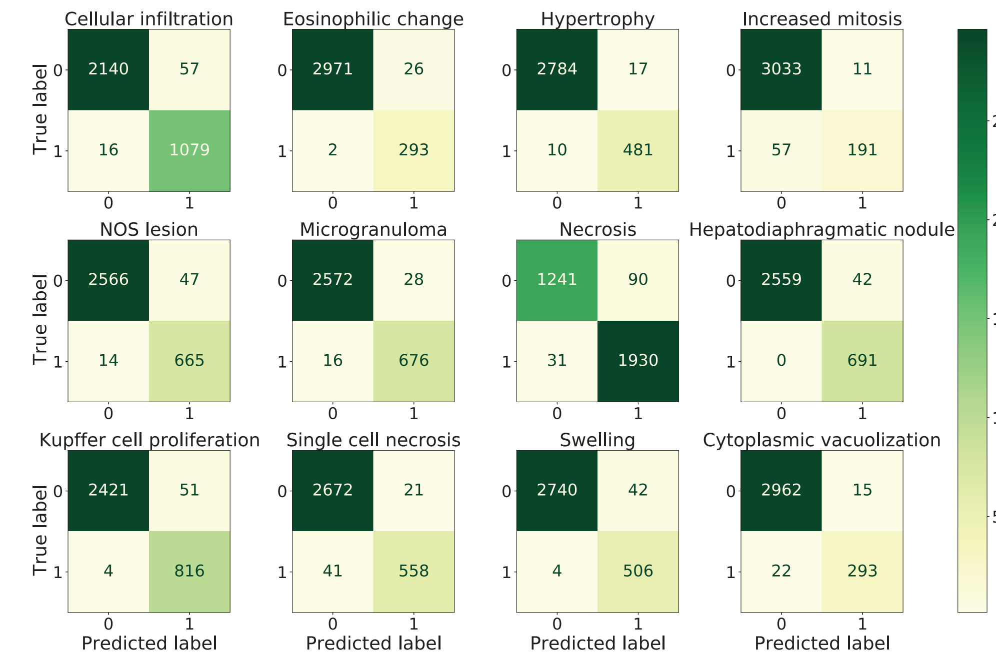
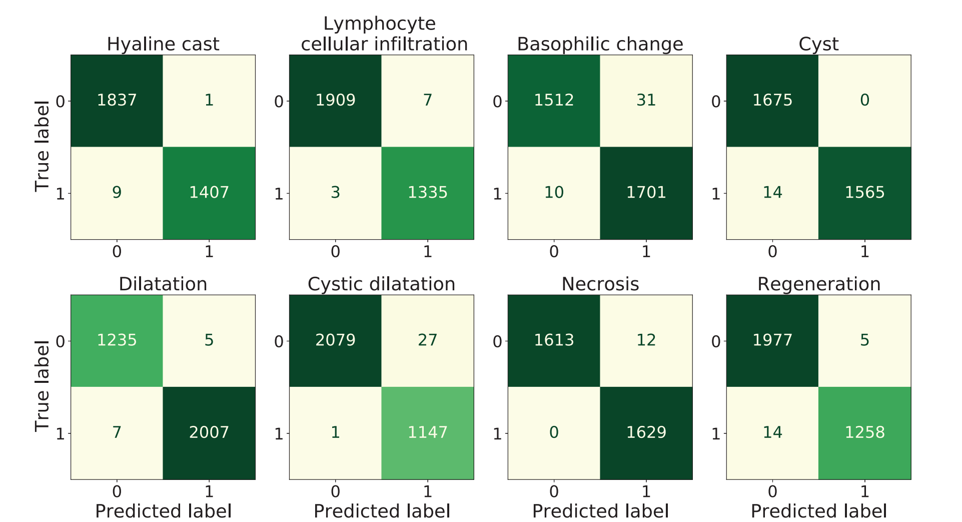
可以为每个标签提供合理的预测准确性。ACC_{lab} 的更直观比较如图5所示。尽管在由LSTM和SRN实现的所提出模型中，每个标签的预测准确性没有显著差异，但与基于SRN的实验相比，我们在RNN层中以LSTM算法作为分类器核心，仍在大多数药物病理发现预测任务中获得了略高的准确率。



(a) Liver



(b) Kidney

Fig 4. The confusion matrix of the pathology classification. The top-left represents the TN, the top-right represents FP, the bottom-left is FN and the bottom-right is TP. (a) shows the confusion matrix on liver data and (b) shows the confusion matrix on kidney data. The results were obtained from the first fold. The results of other folds are shown in S3 Fig.

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can give reasonable prediction accuracy for each label. The more intuitive comparison of ACC_{lab} is shown in Fig 5. Although there is no significant difference in the prediction accuracy of each label between the proposed model implemented by LSTM and SRN, compared with the experiments based on SRN, we used LSTM algorithm as the core of classifier in RNN layer and still obtained a slightly higher accuracy in most tasks of drug pathological findings prediction.

表 1. 基于 LSTM 的 Att-RethinkNet 与 SRN 算法的比较。

ORG[1]	ALG[2]	准确率 (%)	SEN (%)	SPE (%)	F1 (%)	AUC	ACC_{pair}(%)	ACC_{avelab}(%)
肝脏	SRN	87.6	91.9	98.3	92.5	0.9921	90.2	97.4
	LSTM	89.4	94.2	98.2	93.8	0.9930	92.2	97.8
肾脏	SRN	96.4	98.7	99.4	98.9	0.9947	97.6	99.1
	LSTM	97.5	99.1	99.5	99.2	0.9949	98.1	99.3

[1] ORG 意味着目标器官的数据集。[2] ALG 代表在 RNN 层中实际采用的网络结构。

https://doi.org/10.1371/journal.pcbi.1010402.t001

表2. 基于LSTM和SRN的Att-RethinkNet在肝脏数据集上所有病理发现的ACC_{lab}。

ALG/PF[1]	CI	EC	HY	IM	NL	MI	NE	HN	KCP	SCN	SW	CV
SRN	96.2	98.9	98.3	97.4	97.4	97.1	95.3	98.3	96.6	97.3	97.3	98.6
LSTM	96.2	99.0	98.3	98.0	97.6	97.6	96.1	98.6	97.4	97.9	97.7	98.9

[1] ALG 表示 RNN 层中实际采用的网络结构。第 2 列到第 13 列的数值表示对应病理发现的 ACC_{lab} (%)。

https://doi.org/10.1371/journal.pcbi.1010402.t002

从肝脏和肾脏数据集中获得的ROC如图6所示。结果表明，这两种算法在不同的数据集上都获得了较高的AUC值，而且基于LSTM的模型略高于基于SRN的模型。因此，LSTM对历史信息的选择性记忆功能提高了模型识别特定药物是否具有潜在病理发现的能力。

总之，使用LSTM神经网络作为RNN层的具体实现算法来完成药物诱导病理发现的早期识别和分类，其性能比基于SRN的多标签分类模型更令人满意。因此，后续实验中提到的所有Att-RethinkNet模型都是通过在RNN层中使用LSTM算法实现的。

Att-RethinkNet 与 RethinkNet 的比较

本节旨在实现我们提出的 Att-RethinkNet，并将其与基于基因表达数据进行药物诱导病理分类的基线 RethinkNet 框架进行比较。为了公平比较，这两种模型采用相同的数据划分方法和交叉验证程序。两种方法在不同器官中的结果如表 4 所示。根据大多数评估指标，结果显示 Att-RethinkNet 的预测能力强于 RethinkNet。对于鼠肝数据，基线模型的 ACC 达到 87.2%，ACC_{pair} 为 90.2%，而我们提出的模型实现了 ACC 为 89.4%，ACC_{pair} 为 92.2%，SEN 为 94.2%，SPE 为 98.2%，AUC 为 0.99。在肾脏数据方面，Att-RethinkNet 的 ACC 为 97.5%，ACC_{pair} 为 98.1%，SEN 为 99.1%，SPE 为 99.5%，AUC 为 0.99，所有指标均高于 RethinkNet 的结果。

表 3. 基于 LSTM 和 SRN 的 Att-RethinkNet 对肾脏数据集的所有病理发现的 ACC_{lab}。

ALG/PF[1]	HC	LCI	BC	CY	DI	CD	NE	RE
SRN	99.6	99.2	98.8	98.5	99.2	99.3	99.6	98.6
LSTM	99.4	99.4	98.9	99.1	99.5	99.3	99.7	99.1

[1] ALG 代表网络结构操作 通常采用在RNN层。第2到第9列的数值表示对应病理发现的ACC_{lab} (%) ng.

https://doi.org/10.1371/journal.pcbi.1010402.t003

Table 1. Comparison between Att-RethinkNet based on LSTM and SRN algorithms.

ORG[1]	ALG[2]	ACC (%)	SEN (%)	SPE (%)	F1 (%)	AUC	ACC_{pair}(%)	ACC_{avelab}(%)
Liver	SRN	87.6	91.9	98.3	92.5	0.9921	90.2	97.4
	LSTM	89.4	94.2	98.2	93.8	0.9930	92.2	97.8
Kidney	SRN	96.4	98.7	99.4	98.9	0.9947	97.6	99.1
	LSTM	97.5	99.1	99.5	99.2	0.9949	98.1	99.3

[1] ORG means the data set of target organ.
[2] ALG represents the network structure actually adopted in RNN layer.

https://doi.org/10.1371/journal.pcbi.1010402.t001

Table 2. ACC_{lab} of all pathological findings for Att-RethinkNet based on LSTM and SRN for liver data set.

ALG/PF[1]	CI	EC	HY	IM	NL	MI	NE	HN	KCP	SCN	SW	CV
SRN	96.2	98.9	98.3	97.4	97.4	97.1	95.3	98.3	96.6	97.3	97.3	98.6
LSTM	96.2	99.0	98.3	98.0	97.6	97.6	96.1	98.6	97.4	97.9	97.7	98.9

[1] ALG represents the network structure actually adopted in RNN layer. Values of columns 2 to 13 indicate the ACC_{lab} (%) of the corresponding pathological finding.

https://doi.org/10.1371/journal.pcbi.1010402.t002

The ROCs obtained from the liver and kidney data sets are shown in Fig 6. The results show that the two algorithms have obtained high AUC values on different data sets, and the model based on LSTM is slightly higher than the model based on SRN. Therefore, the selective memory function of LSTM for historical information improves the ability of the model to identify whether a specific drug has potential pathological findings.

In a word, using LSTM neural network as the specific implementation algorithm of RNN layer to complete the early recognition and classification of drug-induced pathological findings obtains more satisfactory performance than the multi-label classification model based on SRN. Therefore, all Att-RethinkNet model mentioned in the follow-up experiments was implemented by using LSTM algorithm in RNN layer.

Comparison between Att-RethinkNet and RethinkNet

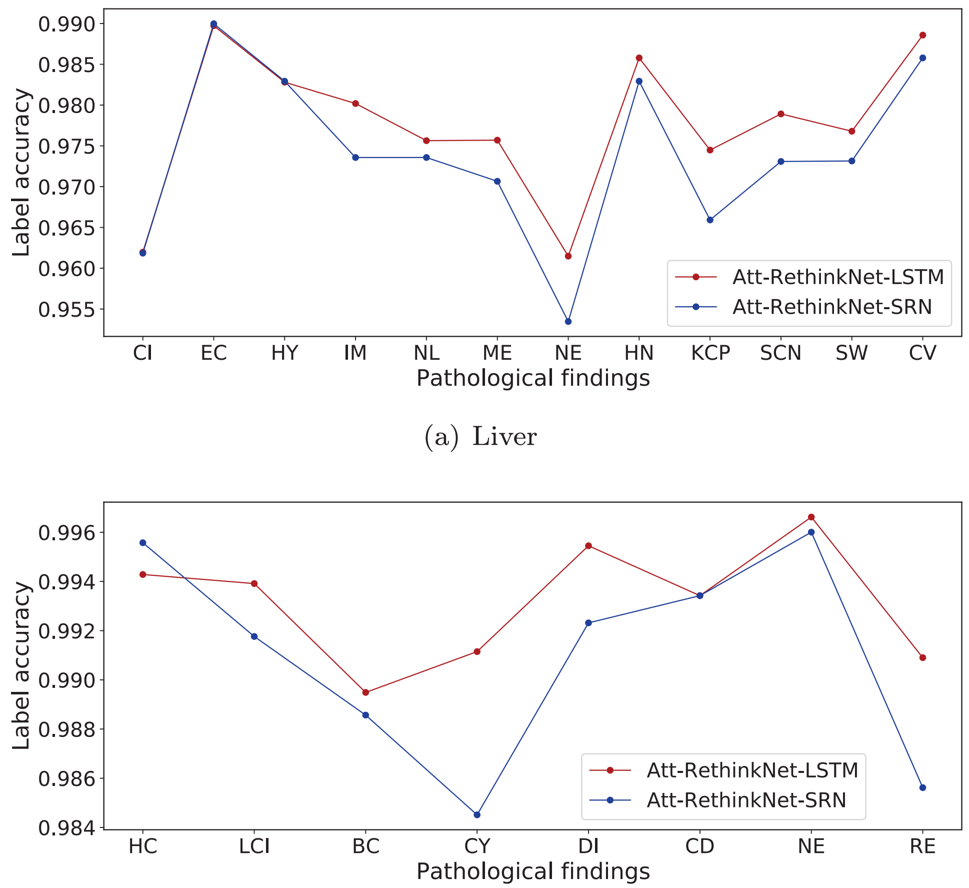
This section aims to implement our proposed Att-RethinkNet and compare it with the baseline RethinkNet framework for drug-induced pathology classification based on gene expression data. For fair comparison, the two models share the same data splitting method and cross validation procedure. The results of the two methods in different organs are shown in Table 4. According to most of the evaluation metrics, it shows that the predictive power of Att-RethinkNet is stronger than that of the RethinkNet. For the rat liver data, the baseline model reached an ACC of 87.2% and an ACC_{pair} of 90.2%, while our proposed model achieved an ACC of 89.4%, an ACC_{pair} of 92.2%, a SEN of 94.2%, a SPE of 98.2% and an AUC of 0.99. In terms of kidney data, Att-RethinkNet has an ACC of 97.5%, an ACC_{pair} of 98.1%, a SEN of 99.1%, a SPE of 99.5% and an AUC of 0.99, which are all higher than the RethinkNet's results. The

Table 3. ACC_{lab} of all pathological findings for Att-RethinkNet based on LSTM and SRN for kidney data set.

ALG/PF[1]	HC	LCI	BC	CY	DI	CD	NE	RE
SRN	99.6	99.2	98.8	98.5	99.2	99.3	99.6	98.6
LSTM	99.4	99.4	98.9	99.1	99.5	99.3	99.7	99.1

[1] ALG represents the network structure actually adopted in RNN layer. Values of columns 2 to 9 indicate the ACC_{lab} (%) of the corresponding pathological finding.

https://doi.org/10.1371/journal.pcbi.1010402.t003

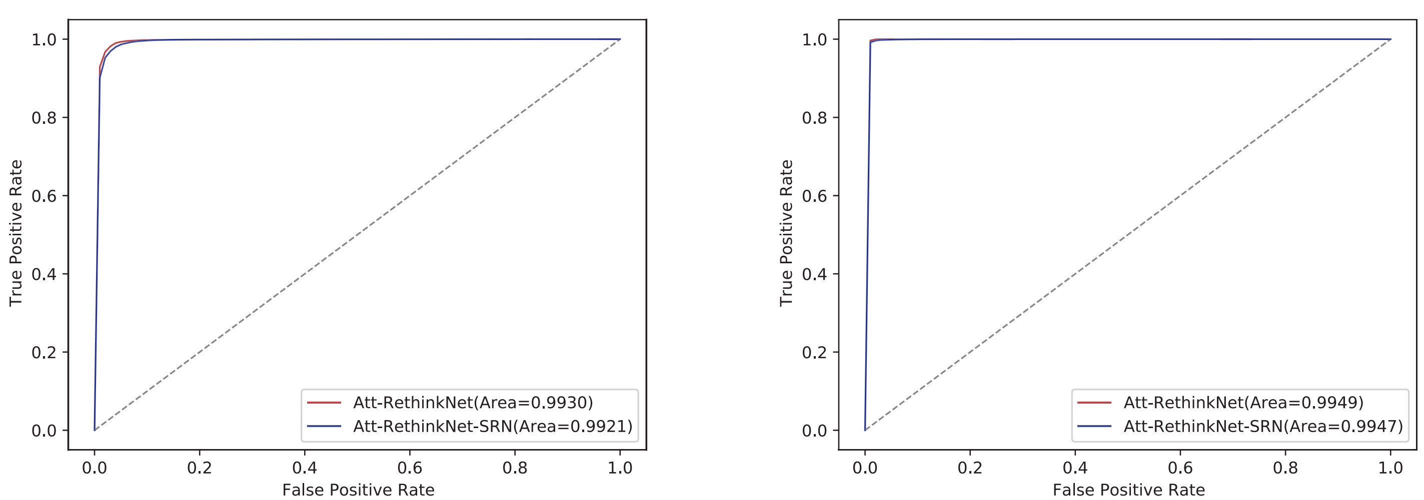


(a) Liver

(b) Kidney

图5. 肝脏数据(a)和肾脏数据(b)深度学习实验的ACC_{lab}。

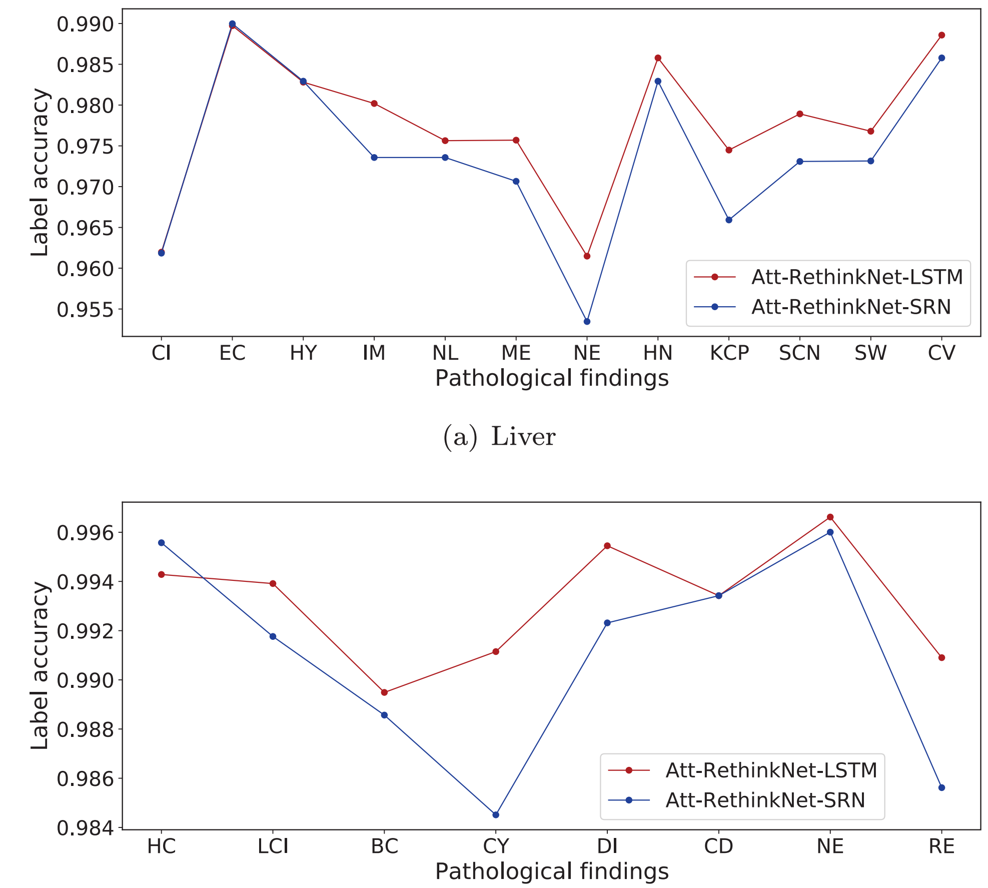
https://doi.org/10.1371/journal.pcbi.1010402.g005



(a) Liver (b) Kidney

图6. 使用LSTM和SRN算法的Att-RethinkNet在肝脏数据(a)和肾脏数据(b)上的ROC曲线。

https://doi.org/10.1371/journal.pcbi.1010402.g006

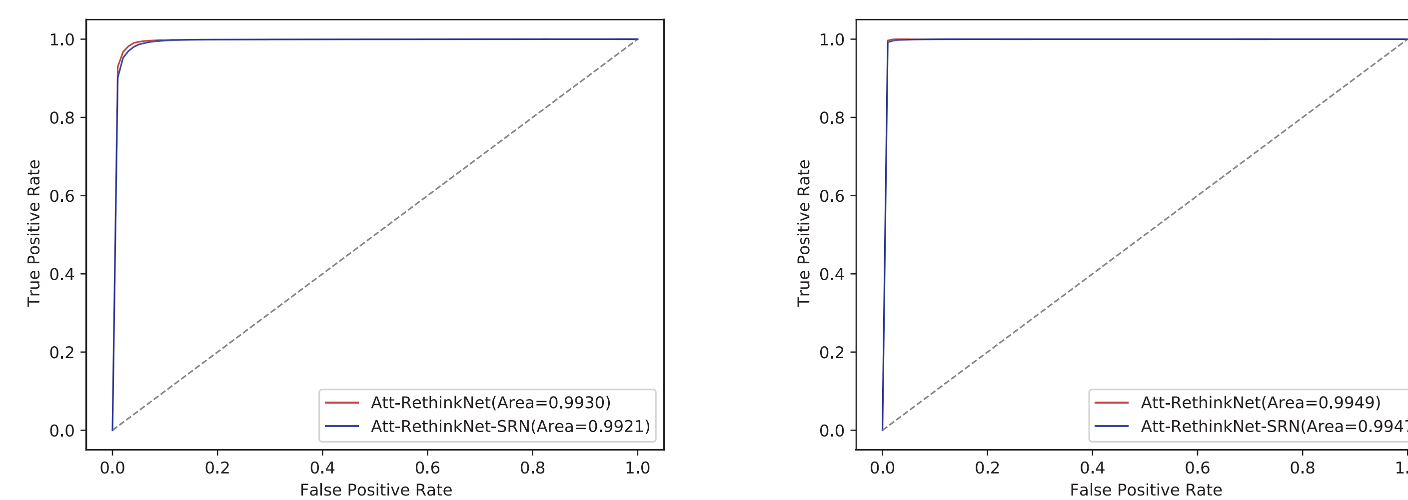


(a) Liver

(b) Kidney

Fig 5. The ACC_{lab} of deep learning experiments in liver data (a) and kidney data (b).

https://doi.org/10.1371/journal.pcbi.1010402.g005



(a) Liver (b) Kidney

Fig 6. ROC curves of Att-RethinkNet using LSTM and SRN algorithms on liver data (a) and kidney data (b).

https://doi.org/10.1371/journal.pcbi.1010402.g006

表 4. C 所提方法与基准方法的比较 模型。

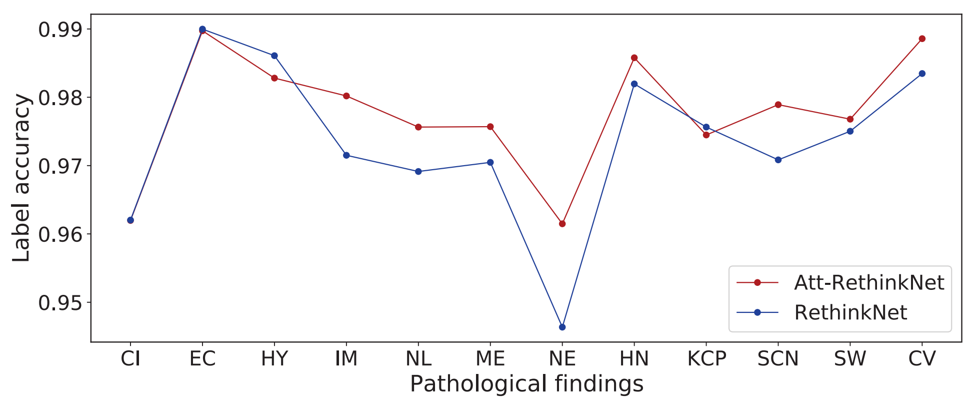
ORG[1]	CLF[2]	准确率 (%)	SEN (%)	SPE (%)	F1 (%)	AUC	ACC_{pair}(%)	ACC_{avelab}(%)
肝脏	再思网	87.2	91.1	98.5	92.3	0.99	90.2	97.4
	Att-RethinkNet	89.4	94.2	98.2	93.8	0.99	92.2	97.8
肾脏	RethinkNet	96.2	98.4	99.5	98.9	0.99	97.5	99.0
	Att-RethinkNet	97.5	99.1	99.5	99.2	0.99	98.1	99.3

[1] ORG 意味着目标器官的数据集。[2]
CLF 意味着分类器。

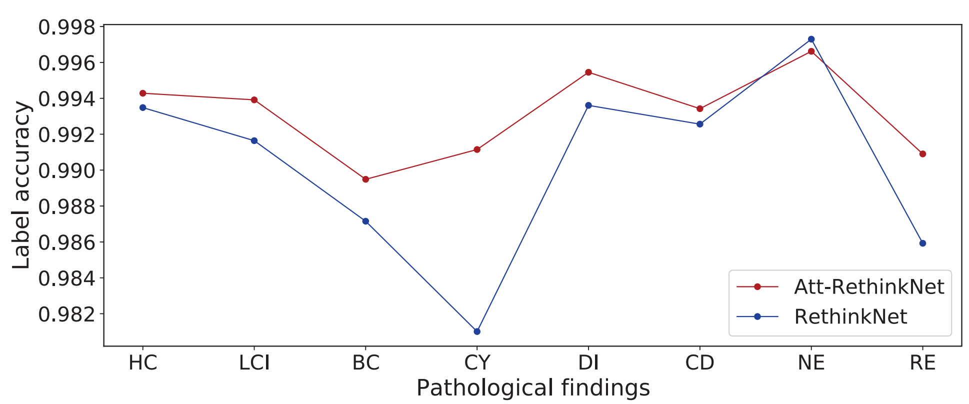
https://doi.org/10.1371/journal.pcbi.1010402.t004

肾脏数据子集准确率高于肝脏数据子集准确率的原因可能是子集准确率是一种严格的度量，也就是说，如果一个样本的标签向量中的一个元素被错误预测，则该样本被视为预测错误。因此，高维标签向量更容易被错误预测。肝脏数据集的标签维度高于肾脏数据集，因此更有可能被判定为预测错误。

为了比较每个标签的分类性能，RethinkNet 和 Att-RethinkNet 的 ACC_{lab} 如图 7 所示。ACC_{lab} 的详细数值汇总在表 5 和表 6 中。如预期，对于大多数标签，其预测都有明显的提升。



(a) Liver



(b) Kidney

图7. 肝脏数据(a)和肾脏数据(b)深度学习实验的ACC_{lab}。

https://doi.org/10.1371/journal.pcbi.1010402.g007

Table 4. Comparison between the proposed method and the baseline model.

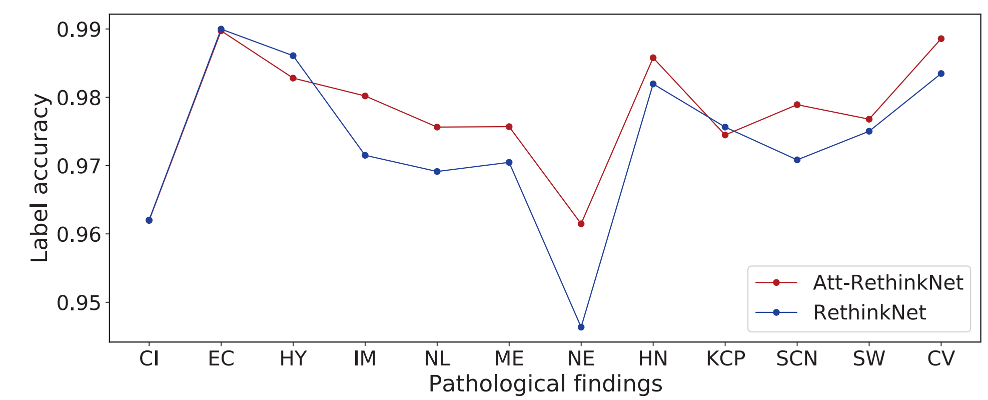
ORG[1]	CLF[2]	ACC (%)	SEN (%)	SPE (%)	F1 (%)	AUC	ACC_{pair}(%)	ACC_{avelab}(%)
Liver	RethinkNet	87.2	91.1	98.5	92.3	0.99	90.2	97.4
	Att-RethinkNet	89.4	94.2	98.2	93.8	0.99	92.2	97.8
Kidney	RethinkNet	96.2	98.4	99.5	98.9	0.99	97.5	99.0
	Att-RethinkNet	97.5	99.1	99.5	99.2	0.99	98.1	99.3

[1] ORG means the data set of target organ.
[2] CLF means classifier.

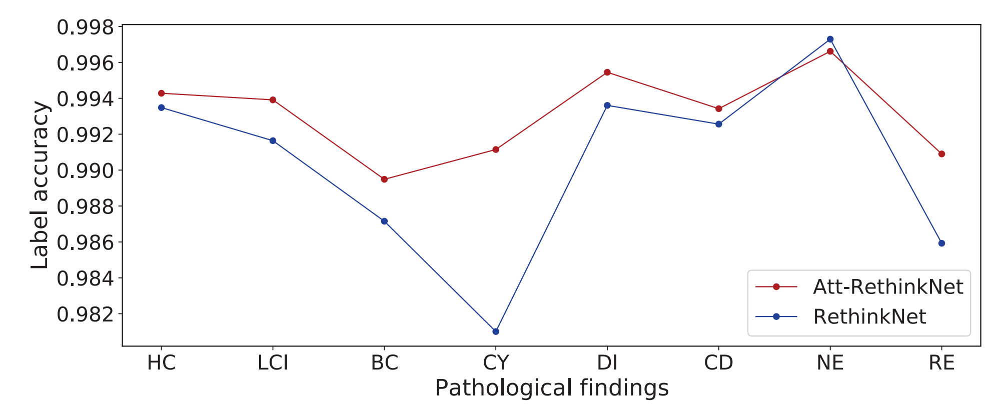
https://doi.org/10.1371/journal.pcbi.1010402.t004

reasons why the subset accuracy of kidney data is higher than that of liver data may be that subset accuracy is a rigid measurement, that is, if one element of one sample's label vector is falsely predicted, the sample is considered falsely predicted. Therefore, high dimensional label vector may be more easily to be falsely predicted. The liver data set has a higher label dimension than that of the kidney, so it is more likely to be judged as a false prediction.

To compare the classification performance on each label, the ACC_{lab} of RethinkNet and Att-RethinkNet is illustrated in Fig 7. The detailed values of ACC_{lab} are summarized in Tables 5 and 6. As expected, obvious improvement of each label's prediction can be seen for most of



(a) Liver



(b) Kidney

Fig 7. The ACC_{lab} of deep learning experiments in liver data (a) and kidney data (b).

https://doi.org/10.1371/journal.pcbi.1010402.g007

表5. RethinkNet和Att-RethinkNet在肝脏数据中所有病理发现的ACC_{lab}。

CLF/PF[1]	CI	EC	HY	IM	NL	MI	NE	HN	KCP	SCN	SW	CV
再思网	96.2	99.0	98.6	97.2	96.9	97.0	94.6	98.2	97.6	97.1	97.5	98.3
Att-RethinkNet	96.2	99.0	98.3	98.0	97.6	97.6	96.1	98.6	97.4	97.9	97.7	98.9

[1] CLF表示分类器，PF表示病理发现。第2到第13列的数值表示对应病理发现的ACC_{lab} (%)。

https://doi.org/10.1371/journal.pcbi.1010402.t005

表6. RethinkNet和Att-RethinkNet在肾脏数据中所有病理发现的ACC_{lab}。

CLF/PF[1]	HC	LCI	BC	CY	DI	CD	NE	RE
再思网	99.3	99.2	98.7	98.1	99.4	99.3	99.7	98.6
Att-RethinkNet	99.4	99.4	98.9	99.1	99.5	99.3	99.7	99.1

[1] CLF 表示分类器，PF 意味着病理发现。第2到第9列的数值表示相应病理发现的ACC_{lab} (%) g.

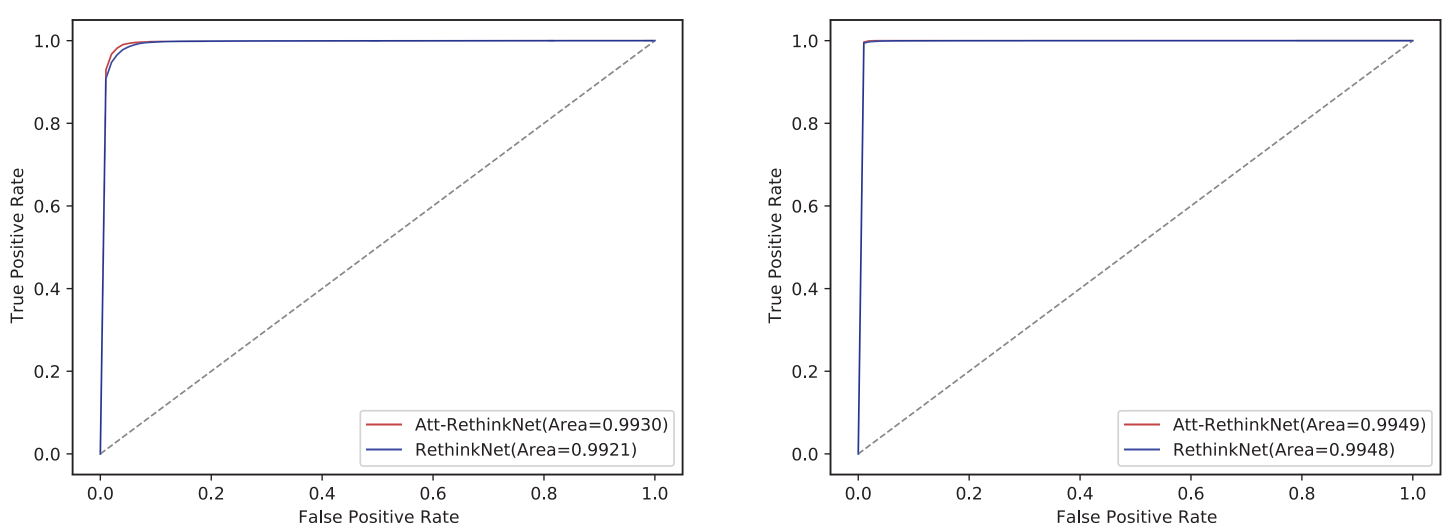
https://doi.org/10.1371/journal.pcbi.1010406

使用我们提出的模型标注的结果，除了四个发现：肝脏的细胞浸润、嗜酸性变化、肥大和库普弗细胞增生，以及一个发现：肾脏的坏死。结果还显示，相较于肝脏的其他发现，肝脏的嗜酸性变化更容易被识别；相较于肾脏的其他发现，肾脏的坏死更容易被识别。

两种方法的ROC曲线如图8所示。根据ROC曲线，Att-RethinkNet的AUC略大于RethinkNet，并且位于RethinkNet的左上方，这意味着我们提出的模型的分类性能优于基线模型。

Att-RethinkNet 与传统方法的比较

传统的分类算法通常在一开始就会降低特征维度并消除无关信息以优化结果。我们应用了一些特征



(a) Liver (b) Kidney

图8. RethinkNet 和 Att-RethinkNet 在肝脏数据 (a) 和肾脏数据 (b) 上的 ROC 曲线。

https://doi.org/10.1371/journal.pcbi.1010402.g008

Table 5. ACC_{lab} of all pathological findings for RethinkNet and Att-RethinkNet in liver data.

CLF/PF[1]	CI	EC	HY	IM	NL	MI	NE	HN	KCP	SCN	SW	CV
RethinkNet	96.2	99.0	98.6	97.2	96.9	97.0	94.6	98.2	97.6	97.1	97.5	98.3
Att-RethinkNet	96.2	99.0	98.3	98.0	97.6	97.6	96.1	98.6	97.4	97.9	97.7	98.9

[1] CLF means classifier and PF means pathological finding. Values of columns 2 to 13 indicate the ACC_{lab} (%) of the corresponding pathological finding.

https://doi.org/10.1371/journal.pcbi.1010402.t005

Table 6. ACC_{lab} of all pathological findings for RethinkNet and Att-RethinkNet in kidney data.

CLF/PF[1]	HC	LCI	BC	CY	DI	CD	NE	RE
RethinkNet	99.3	99.2	98.7	98.1	99.4	99.3	99.7	98.6
Att-RethinkNet	99.4	99.4	98.9	99.1	99.5	99.3	99.7	99.1

[1] CLF represents classifier and PF means pathological finding. Values of columns 2 to 9 indicate the ACC_{lab} (%) of the corresponding pathological finding.

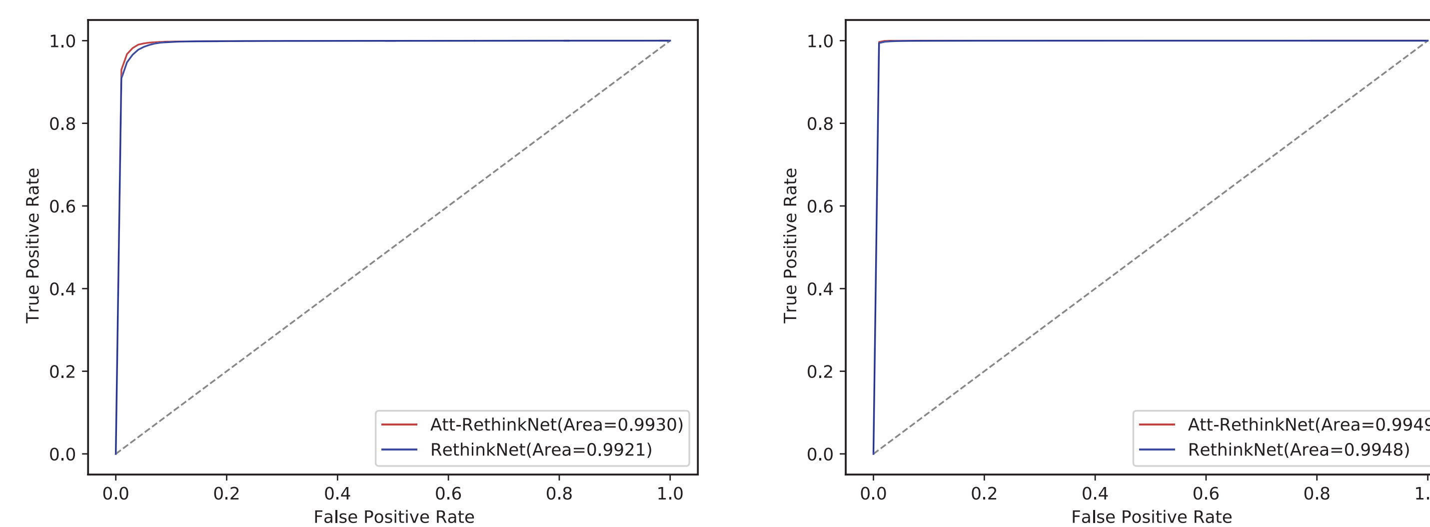
https://doi.org/10.1371/journal.pcbi.1010402.t006

the labels with our proposed model, except four findings, cellular infiltration, eosinophilic change, hypertrophy and kupffer cell proliferation in liver, and one finding, necrosis, in kidney. It also shows that eosinophilic change in liver is easier to be recognized compared with other findings for liver and necrosis in kidney is easier to be identified compared with other findings in kidney.

The ROCs of both methods are shown in Fig 8. According to the ROCs, Att-RethinkNet has a slightly larger AUC than that of RethinkNet and lies in the left-top of RethinkNet, meaning that our proposed model has a better classification performance than the baseline model.

Comparison between Att-RethinkNet and the traditional method

Traditional classification algorithms normally reduce the feature dimension and eliminate irrelevant information to optimize the results at the beginning. We applied some feature



(a) Liver (b) Kidney

Fig 8. ROC curves of RethinkNet and Att-RethinkNet on liver data (a) and kidney data (b).

https://doi.org/10.1371/journal.pcbi.1010402.g008

排名技术发现，多标签F统计算法在特征子集上表现出更好且更稳定的准确性。因此，我们计算了每个基因的F统计得分，并通过逐步删除排名特征选择表现最好的TOP_N个基因。我们使用适应度函数评估每个特征子集的性能 [36]。由于所选子集中的特征数量明显少于所有特征的总数，我们通过添加放大因子 λ改进了适应度函数，以最大化分类准确性并最小化所选基因数量。在适应度函数中，我们将所选特征子集增加到λ倍，以使适应度值具有意义。改进的适应度函数定义如下：

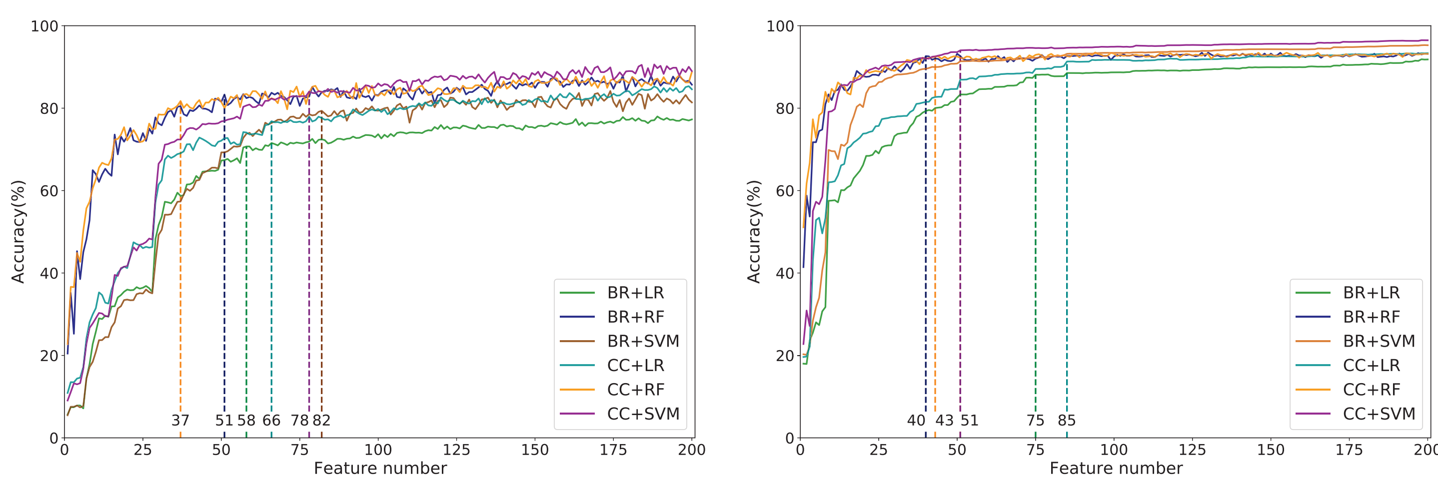
$$\text{Fitness} = \alpha \times \text{ACC} + (1-\alpha) \times \frac{D_{total} - D_{selected} \times \lambda}{D_{total}} \tag{13}$$

其中 ACC 是准确率。D_{total} 和 $D_{selected}$ 分别表示总特征的大小和所选特征的大小。 α 是一个在 [0, 1]，范围内的权重，用于描述 ACC 和 $D_{selected}$ 的重要程度。 λ 是一个放大因子。在我们的实验中，我们尝试了多组参数，最终设定为 $\alpha = 0.6$ 和 $\lambda = 10$，这组参数具有最高的 ACC。

在本文中，我们应用了改进的适应度函数以寻求相关特征的最优子集。选择最优特征子集的中间结果如图9所示，其中x轴表示用于机器学习模型构建的所选特征数量，y轴表示使用所选特征集对未知样本进行分类时的子集准确率。在这里，我们将BR/CC与LR、RF和SVM结合使用，它们都是相关领域中流行的分类器 [37–41]。

从图9来看，对于肝脏数据，基于SVM的BR选择了最多的特征，而基于RF的CC选择的特征最少，大约只有其他方法的一半。基于SVM的CC达到了最高的准确性。对于肾脏数据的特征选择，基于SVM的BR和CC方法选择了相同数量的基因，基于SVM的CC达到了最高的准确性。

我们在表7中进一步展示了BR、CC和Att-RethinkNet的分类比较。从表7可以看出，传统机器学习模型对肝脏数据的预测准确率范围为69.86%到83.71%，对肾脏数据的预测准确率为88.74%到94.66%。



(a) Liver　　(b) Kidney

图 9. 中间结果　　选择 BR 和 CC 的最佳特征子集的方法。选定特征的数量用虚线标出　　es.

https://doi.org/10.1371/journal.pcbi.1010402.g009

ranking techniques and found that multi-label F-statistic algorithm show better and more stable accuracy in feature subset. So we calculated the F-statistic score of each gene and picked the TOP_N-best performed genes by deleting ranked features gradually. We used a fitness function to evaluate the performance of each feature subset [36]. Since the number of features in the selected subset is significantly smaller than the number of all features, we improved the fitness function by adding an amplification factor λ in order to maximizes the accuracy of classification and minimizes the number of selected genes. In the fitness function, we increased the selected feature subset to λ times to make the fitness value meaningful. The improved fitness function is defined as:

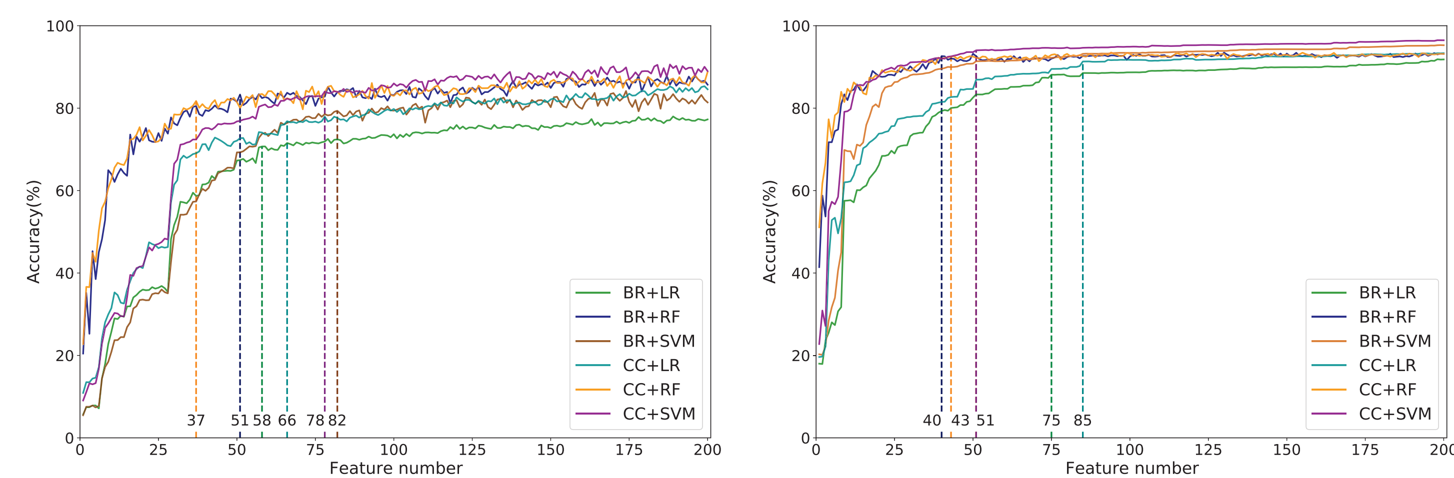
$$\text{Fitness} = \alpha \times \text{ACC} + (1-\alpha) \times \frac{D_{total} - D_{selected} \times \lambda}{D_{total}} \tag{13}$$

Where ACC is the accuracy. D_{total} and $D_{selected}$ represent the size of the total features and the size of the selected features, respectively. α is a weight in the range [0, 1], which describes the degree of importance of ACC and $D_{selected}$. λ is an amplification factor. In our experiments, we tried multiple sets of parameters and finally set $\alpha = 0.6$ and $\lambda = 10$ which had the highest ACC.

In this paper, we applied the improved fitness function to seek an optimal subset of relevant features. The intermediate results of selecting the optimal feature subset are presented in Fig 9, where x-axis shows the number of selected features that were used for machine learning model construction and y-axis shows the subset accuracy when classifying unknown samples using the selected feature sets. Here we combined the BR/CC with LR, RF and SVM, which are all popular classifiers in relevant areas [37–41].

From Fig 9, for liver data, BR based on SVM selected the most features, and CC based on RF selected the least features, approximately only one-half of the other methods. CC based on SVM achieved the highest accuracy. For feature selection in kidney data, BR and CC methods based on SVM selected the same number of genes and CC based SVM achieved the highest accuracy.

We further show the classification comparison of BR, CC and Att-RethinkNet in Table 7. From Table 7, we found that the prediction accuracy of traditional machine learning-based models for liver data ranges from 69.86% to 83.71% and for kidney data, 88.74% to 94.66%.



(a) Liver　　(b) Kidney

Fig 9. Intermediate results of selecting the optimal feature subset for BR and CC. The numbers of selected features are marked with dashed lines.

https://doi.org/10.1371/journal.pcbi.1010402.g009

表7. BR、CC和Att-RethinkNet在肝脏和肾脏数据上的性能。

ORG[1]	CLF[2]	BCLF[3]	FN[4]	准确率 (%)	SEN (%)	SPE (%)	F1 (%)	AUC	ACC_{pair} (%)	ACC_{avelab} (%)
肝脏	BR	LR	58	69.9	85.3	96.5	87.3	0.98	78.3	95.5
	BR	RF	51	81.5	92.0	98.3	95.0	0.99	87.2	98.0
	BR	SVM	82	78.7	88.7	97.0	89.5	0.99	84.1	96.5
	CC	LR	66	76.0	84.0	96.4	85.4	0.96	79.2	95.0
	CC	RF	37	79.6	89.6	98.2	93.4	0.99	85.2	97.6
	CC	SVM	78	83.7	88.0	97.0	88.5	0.97	85.5	96.1
	Att-RethinkNet	-	-	89.4	94.2	98.2	93.8	0.99	92.2	97.8
肾脏	BR	LR	75	88.7	97.4	95.6	96.3	0.99	92.1	96.6
	BR	RF	40	92.6	99.5	97.0	98.2	0.99	93.9	98.3
	BR	SVM	51	91.8	98.9	96.5	97.5	0.99	93.9	97.7
	CC	LR	85	91.8	98.5	96.2	97.2	0.99	92.9	97.4
	CC	RF	43	92.0	99.2	97.4	98.2	0.99	93.9	98.3
	CC	SVM	51	94.7	99.4	97.4	98.2	0.99	95.5	98.3
	Att-RethinkNet	-	-	97.5	99.1	99.5	99.2	0.99	98.1	99.3

[1] ORG 意味着目标器官的数据。[2] CLF 意味着分类器。[3] BCLF 意味着基础分类器。[4] FN 意味着选定的特征子集大小。

https://doi.org/10.1371/journal.pcbi.1010402.t007

在这些方法中，可以看出使用LR作为基分类器的BR在两个数据集上的准确率最低。使用RF作为基分类器的BR和CC在所有数据集上获得了最佳的AUC。在CC方法中，当SVM用作基分类器时，分类器始终具有最高的准确率，肝脏数据为83.71%，肾脏数据为94.66%。总体而言，所有这些基于机器学习的方法都保留了相当小的特征子集，而且肾脏数据的结果优于肝脏数据。此外，在使用相同基分类器的情况下，CC在大多数情况下优于BR，因为CC考虑了标签之间的相关性。在Att-RethinkNet方面，Att-RethinkNet在肝脏和肾脏数据上都实现了最高的ACC、ACC_{pair} 和ACC_{avelab}。一个重要原因是我们的模型不仅考虑了标签相关性，还对标签和特征都应用了适当的权重，并解决了由标签顺序引起的问题。

此外，每个标签在 Att-RethinkNet 和这些基于机器学习的模型中的详细 ACC_{lab} 如图 10、表 8 和表 9 所示。对于肝脏病理发现，与其他方法相比，Att-RethinkNet 的平均值保持较高水平。对于肾脏病理发现，Att-RethinkNet 每个标签的 ACC_{lab} 在与所有其他传统分类方法的比较中最高。

我们在图11中展示了Att-RethinkNet和所有传统模型的ROC曲线。我们在所有方法中获得了Att-RethinkNet的最高AUC值。

Att-RethinkNet与综合模型的比较

我们还将所提出的方法与在我们的研究中称为“综合模型”的方法进行了比较 [22]。该模型也是为药物诱导的病理发现预测而开发的。不同于我们的方法构建一个多标签预测模型，该模型为每个病理发现训练一个模型，并将所有病理预测模型结合起来。

Table 7. The performance of BR, CC and Att-RethinkNet on liver and kidney data.

ORG[1]	CLF[2]	BCLF[3]	FN[4]	ACC (%)	SEN (%)	SPE (%)	F1 (%)	AUC	ACC_{pair} (%)	ACC_{avelab} (%)
Liver	BR	LR	58	69.9	85.3	96.5	87.3	0.98	78.3	95.5
	BR	RF	51	81.5	92.0	98.3	95.0	0.99	87.2	98.0
	BR	SVM	82	78.7	88.7	97.0	89.5	0.99	84.1	96.5
	CC	LR	66	76.0	84.0	96.4	85.4	0.96	79.2	95.0
	CC	RF	37	79.6	89.6	98.2	93.4	0.99	85.2	97.6
	CC	SVM	78	83.7	88.0	97.0	88.5	0.97	85.5	96.1
	Att-RethinkNet	-	-	89.4	94.2	98.2	93.8	0.99	92.2	97.8
Kidney	BR	LR	75	88.7	97.4	95.6	96.3	0.99	92.1	96.6
	BR	RF	40	92.6	99.5	97.0	98.2	0.99	93.9	98.3
	BR	SVM	51	91.8	98.9	96.5	97.5	0.99	93.9	97.7
	CC	LR	85	91.8	98.5	96.2	97.2	0.99	92.9	97.4
	CC	RF	43	92.0	99.2	97.4	98.2	0.99	93.9	98.3
	CC	SVM	51	94.7	99.4	97.4	98.2	0.99	95.5	98.3
	Att-RethinkNet	-	-	97.5	99.1	99.5	99.2	0.99	98.1	99.3

[1] ORG means the data of target organ.
[2] CLF means classifier.
[3] BCLF means base classifier.
[4] FN means selected feature subset size.

https://doi.org/10.1371/journal.pcbi.1010402.t007

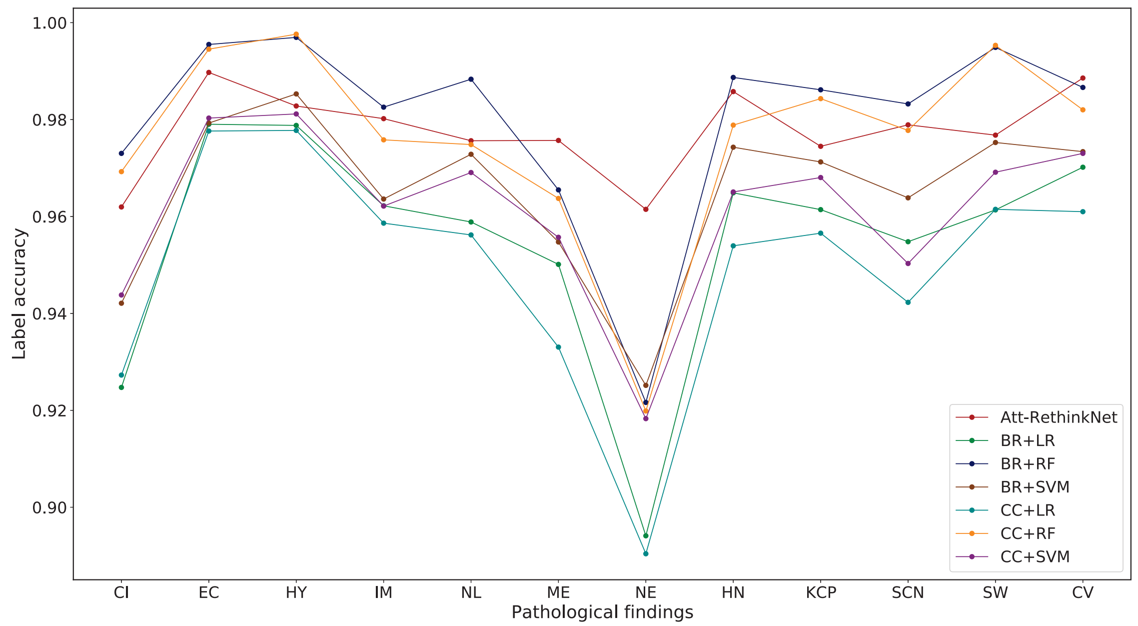
Among these methods, it can be seen that BR using LR as base classifier has the lowest accuracy on both data sets. BR and CC, with RF as the base classifiers, gained the best AUCs in all data sets. In CC approach, when SVM was used as a base classifier, the classifiers always come with the highest accuracy score, 83.71% in liver and 94.66% in kidney respectively. In general, all these machine learning-based methods retain considerable small feature subsets and the result on kidney data is better than that on the liver data. Besides, when using the same base classifiers, CC outperforms BR in most times because CC takes label correlations into account. In terms of the Att-RethinkNet, the Att-RethinkNet achieves the highest ACC, ACC_{pair} and ACC_{avelab} for both liver and kidney data. One important reason is that our model not only considers label correlations but also applies proper weights to both labels and features, and solves the issue caused by label order as well.

Furthermore, the detailed ACC_{lab} of each label for the Att-RethinkNet and these machine learning-based models are shown in Fig 10, Tables 8 and 9. For liver pathological findings, the Att-RethinkNet maintains a high value on average compared with other methods. For kidney pathological findings, the ACC_{lab} of each label of Att-RethinkNet is the highest in comparison with all other referring traditional classification methods.

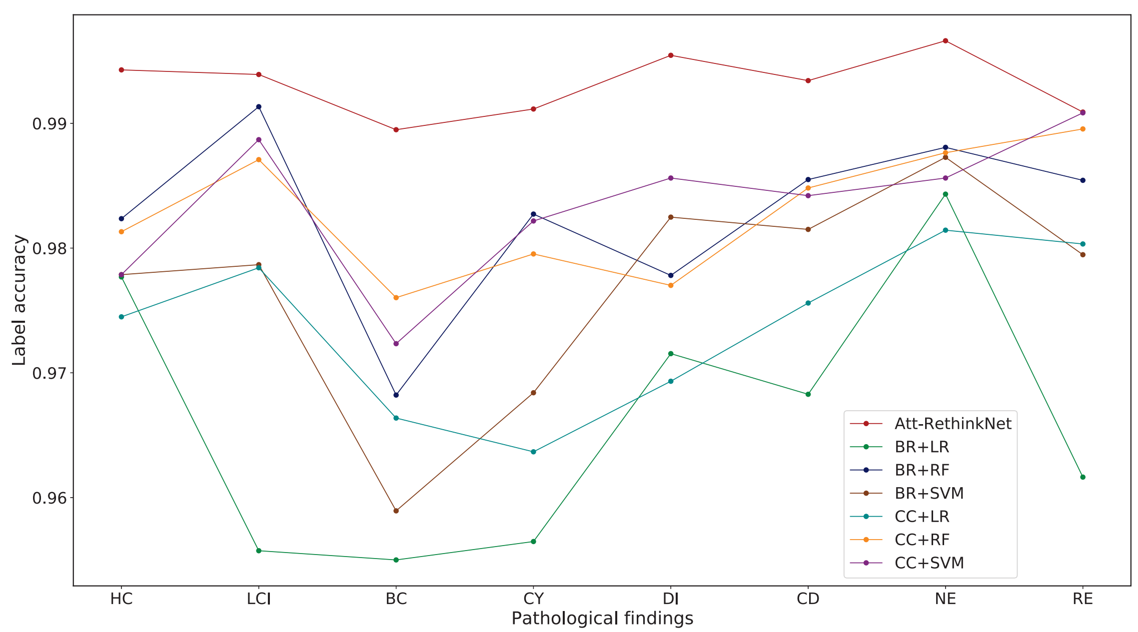
We show the ROC curves of Att-RethinkNet and all the traditional models in Fig 11. We obtain the highest AUC values of the AttRethinkNet among all the methods.

Comparison between Att-RethinkNet and the integrative model

We also compared the proposed approach with a method, that we called "integrative model" in our study [22]. This model was also developed for drug-induced pathological finding prediction. Different from our method, which builds a multi-label prediction model, this model trains a model for each pathological finding and combined all the pathology prediction models.



(a) Liver

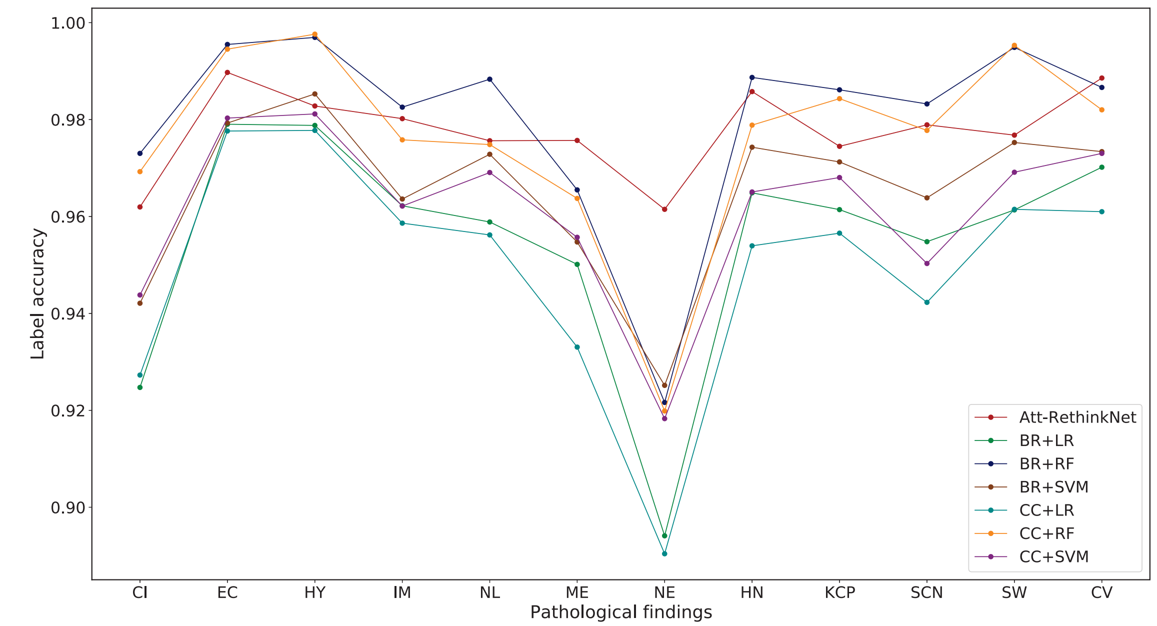


(b) Kidney

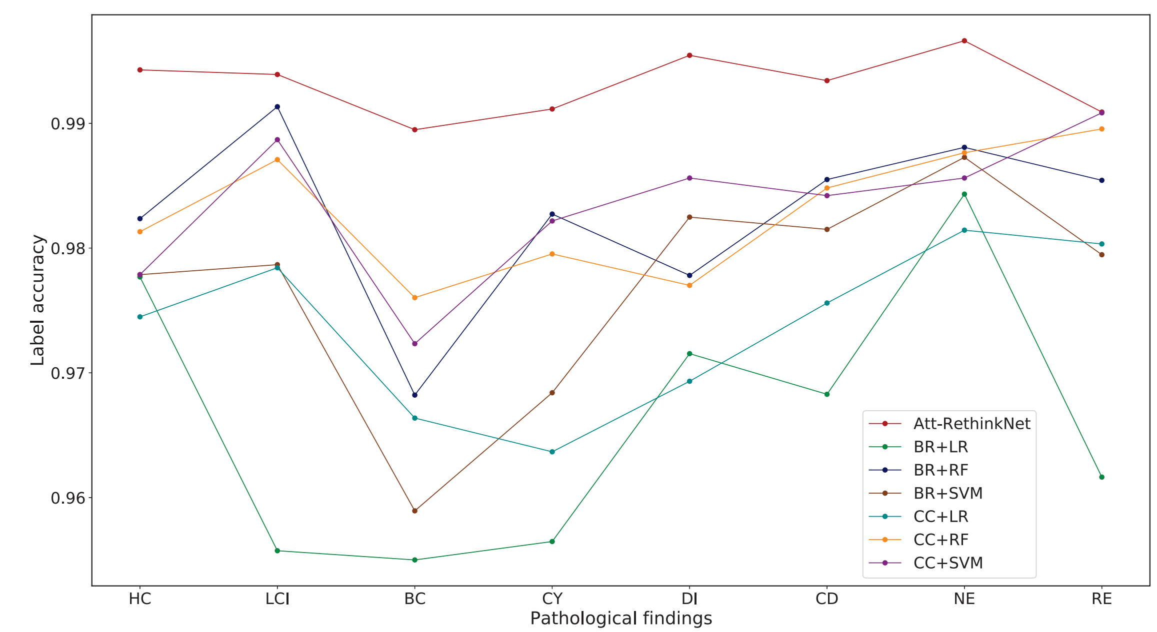
图10. A Att-RethinkNet 和传统基于机器学习的方法在肝脏 (a) 和肾脏 每个标签的准确性 y (b).

https://doi.org/10.1371/journal.pcbi.1010402.g010

我们训练了所展示的5近邻分类器的整合模型。描述训练集中每个折叠内两种病理发现共现情况的病理相似性矩阵在S4图中有报告。表10、11和12列出了整合模型和所提出的药物毒性预测模型的性能。从这些表中，我们可以看到



(a) Liver



(b) Kidney

Fig 10. Accuracy of each label of Att-RethinkNet and the traditional machine learning-based methods in liver (a) and kidney (b).

https://doi.org/10.1371/journal.pcbi.1010402.g010

We trained the presented integrative model of 5-nearest neighbor classifiers. The pathology similarities matrix that describes co-occurrences of two pathological findings within training set of each fold were reported in S4 Fig. Tables 10, 11 and 12 list the performance of the integrative model and the proposed drug toxicity prediction model. From the tables, we can see

表8. BR、CC和Att-RethinkNet在肝脏数据中所有病理发现的表现。

CLF/PF[1]	BCLF[2]	CI	EC	HY	IM	NL	MI	NE	HN	KCP	SCN	SW	CV
BR	LR	92.5	97.9	97.9	96.2	95.9	95.0	89.4	96.5	96.1	95.5	96.1	97.0
BR	RF	97.3	99.6	99.7	98.3	98.8	96.5	92.2	98.9	98.6	98.3	99.5	98.7
BR	SVM	94.2	97.9	98.5	96.4	97.3	95.5	92.5	97.4	97.1	96.4	97.5	97.3
CC	LR	92.7	97.8	97.8	95.9	95.6	93.3	89.0	95.4	95.7	94.2	96.1	96.1
CC	RF	96.9	99.5	99.8	97.6	97.5	96.4	92.0	97.9	98.4	97.8	99.5	98.2
CC	SVM	94.4	98.0	98.1	96.2	96.9	95.6	91.8	96.5	96.8	95.0	96.9	97.3
Att-RethinkNet	-	96.2	99.0	98.3	98.0	97.6	97.6	96.1	98.6	97.4	97.9	97.7	98.9

[1] CLF代表分类器，PF表示病理发现。[2] BCLF表示基础分类器。第3到第14列表示相应病理发现的ACC(%)。

https://doi.org/10.1371/journal.pcbi.1010402.t008

表9. 已建立深度学习模型在肾脏数据中所有病理发现的准确性。

CLF/PF[1]	BCLF[2]	HC	LCI	BC	CY	DI	CD	NE	RE
BR	LR	97.8	95.6	95.5	95.6	97.2	96.8	98.4	96.2
BR	RF	98.2	99.1	96.8	98.3	97.8	98.5	98.8	98.5
BR	SVM	97.8	97.9	95.9	96.8	98.2	98.1	98.7	97.9
CC	LR	97.4	97.8	96.6	96.4	96.9	97.6	98.1	98.0
CC	RF	98.1	98.7	97.6	98.0	97.7	98.5	98.8	99.0
CC	SVM	97.8	98.9	97.2	98.2	98.6	98.4	98.6	99.1
Att-RethinkNet	-	99.4	99.4	98.9	99.1	99.5	99.3	99.7	99.1

[1] CLF代表分类器，PF表示病理发现。[2] BCLF表示基础分类器。第3到第10列表示相应病理发现的ACC(%)。

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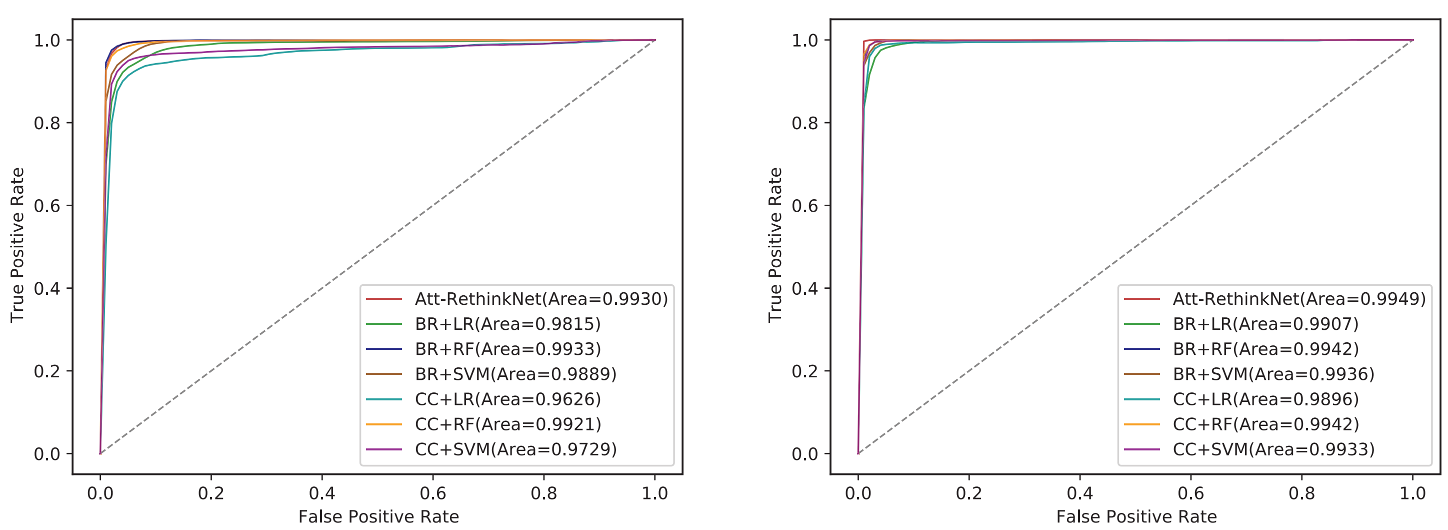

图11. 提出方法与一些常用基于机器学习的模型的ROC曲线比较。(a)显示肝脏数据，(b)显示肾脏数据。在图中，BR加LR表示使用LR作为基分类器的BR模型。其他符号定义类似。

https://doi.org/10.1371/journal.pcbi.1010402.g011

Table 8. Performance of all pathological findings for BR, CC and Att-RethinkNet in liver data.

CLF/PF[1]	BCLF[2]	CI	EC	HY	IM	NL	MI	NE	HN	KCP	SCN	SW	CV
BR	LR	92.5	97.9	97.9	96.2	95.9	95.0	89.4	96.5	96.1	95.5	96.1	97.0
BR	RF	97.3	99.6	99.7	98.3	98.8	96.5	92.2	98.9	98.6	98.3	99.5	98.7
BR	SVM	94.2	97.9	98.5	96.4	97.3	95.5	92.5	97.4	97.1	96.4	97.5	97.3
CC	LR	92.7	97.8	97.8	95.9	95.6	93.3	89.0	95.4	95.7	94.2	96.1	96.1
CC	RF	96.9	99.5	99.8	97.6	97.5	96.4	92.0	97.9	98.4	97.8	99.5	98.2
CC	SVM	94.4	98.0	98.1	96.2	96.9	95.6	91.8	96.5	96.8	95.0	96.9	97.3
Att-RethinkNet	-	96.2	99.0	98.3	98.0	97.6	97.6	96.1	98.6	97.4	97.9	97.7	98.9

[1] CLF represents classifier and PF means pathological finding.
[2] BCLF means base classifier. Columns 3 to 14 indicate the ACC_{lab} (%) of the corresponding pathological finding.

https://doi.org/10.1371/journal.pcbi.1010402.t008

Table 9. Accuracy of all pathological findings for established deep learning models in kidney data.

CLF/PF[1]	BCLF[2]	HC	LCI	BC	CY	DI	CD	NE	RE
BR	LR	97.8	95.6	95.5	95.6	97.2	96.8	98.4	96.2
BR	RF	98.2	99.1	96.8	98.3	97.8	98.5	98.8	98.5
BR	SVM	97.8	97.9	95.9	96.8	98.2	98.1	98.7	97.9
CC	LR	97.4	97.8	96.6	96.4	96.9	97.6	98.1	98.0
CC	RF	98.1	98.7	97.6	98.0	97.7	98.5	98.8	99.0
CC	SVM	97.8	98.9	97.2	98.2	98.6	98.4	98.6	99.1
Att-RethinkNet	-	99.4	99.4	98.9	99.1	99.5	99.3	99.7	99.1

[1] CLF represents classifier and PF means pathological finding.
[2] BCLF means base classifier. Columns 3 to 10 indicate the ACC_{lab} (%) of the corresponding pathological finding.

https://doi.org/10.1371/journal.pcbi.1010402.t009

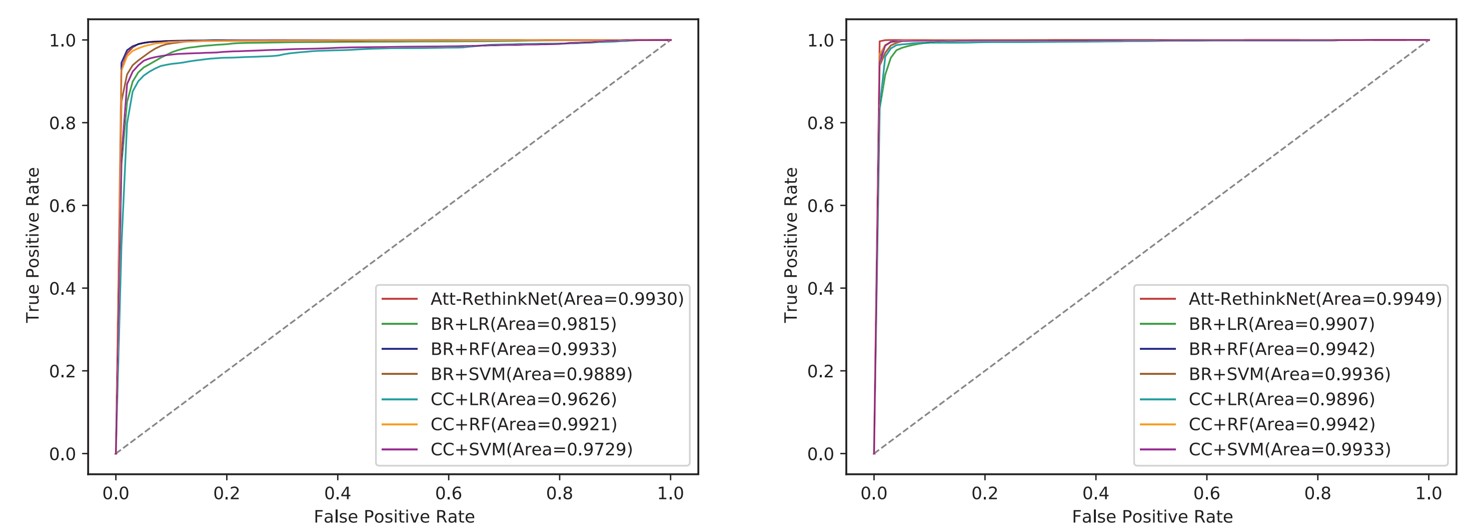

Fig 11. The comparison of ROC curves between the proposed method and some commonly used machine learning-based models. (a) shows the liver data and (b) shows the kidney data. In the plot, BR plus LR represents BR model that uses LR as base classifier. Other symbols are defined similarly.

https://doi.org/10.1371/journal.pcbi.1010402.g011

表 10. 综合模型和 Att-RethinkNet 的性能。

ORG[1]	CLF[2]	准确率 (%)	SEN (%)	SPE (%)	F1 (%)	AUC	ACC_{pair} (%)	ACC_{avelab} (%)
肝脏	Integra	50.3	99.3	89.8	84.9	0.50	83.0	92.6
	Att	89.4	94.2	98.2	93.8	0.99	92.2	97.8
肾脏	Integra	28.5	100.0	41.0	74.2	0.50	63.8	68.4
	Att	97.5	99.1	99.5	99.2	0.99	98.1	99.3

[1] ORG表示目标器官的数据集。[2] CLF表示分类器。Integra表示整合模型。Att表示Att-RethinkNet。

https://doi.org/10.1371/journal.pcbi.1010402.t010

表 11. 肝脏数据的综合模型 ACC_{lab} 与 Att-RethinkNet。

CLF[1]	CI	EC	HY	IM	NL	MI	NE	HN	KCP	SCN	SW	CV
整合模型	83.3	95.8	90.1	98.5	88.2	86.6	91.9	90.0	95.0	97.9	95.0	98.7
Att-RethinkNet	96.2	99.0	98.3	98.0	97.6	97.6	96.1	98.6	97.4	97.9	97.7	98.9

[1] CLF 意味着分类器。第2列到第13列表示相应病理发现的 ACC_{lab} (%)。

https://doi.org/10.1371/journal.pcbi.1010402.t011

尽管综合模型在对每个标签进行分类时获得了相当高的分类精度（虽然低于所提出的方法，如表11和表12所示），但综合模型的子集精度不令人满意（表10）。子集精度低的原因在于综合模型是对每个标签分别进行预测，这无法保证每种病理的预测结果同时正确。

在预测每个标签方面，我们在图12中专门展示了所提出模型和综合模型的ACC_{lab}。结果证明，与综合方法相比，所提出的模型在正确预测每种病理中有显著的提高。对于药物引起的肝毒性（除了增殖分裂(IM)和单细胞坏死(SCN)），我们的方法与综合方法的ACC_{lab}差异大约从1%到12%不等，而对于药物引起的肾毒性，ACC_{lab}差异范围为20%到42%。

Att-RethinkNet 和集成模型的 ROC 曲线可以在图 13 中找到。从实验结果可以看出，Att-RethinkNet 的曲线远高于集成模型的曲线。集成模型的 AUC 大约是 Att-RethinkNet 的一半。

对大鼠肝脏体外数据的验证

为了进一步验证预测模型的可靠性，我们在独立且不可见的数据上进行了实验，即大鼠肝脏的体外毒性数据。我们将数据分为两部分。一部分数据用作训练集，并对其进行了增强和平衡，

表12. 肾脏数据的整合模型ACC_{lab} 与Att-RethinkNet。

CLF[1]	HC	LCI	BC	CY	DI	CD	NE	RE
整合模型	56.8	68.0	79.1	64.7	73.4	62.2	76.8	66.0
Att-RethinkNet	99.4	99.4	98.9	99.1	99.5	99.3	99.7	99.1

[1] CLF 表示分类器。第 2 列到第 9 列表示相应病理发现的 ACC_{lab} (%)。

https://doi.org/10.1371/journal.pcbi.1010402.t012

Table 10. The performance of the integrative model and Att-RethinkNet.

ORG[1]	CLF[2]	ACC (%)	SEN (%)	SPE (%)	F1 (%)	AUC	ACC_{pair} (%)	ACC_{avelab} (%)
Liver	Integra	50.3	99.3	89.8	84.9	0.50	83.0	92.6
	Att	89.4	94.2	98.2	93.8	0.99	92.2	97.8
Kidney	Integra	28.5	100.0	41.0	74.2	0.50	63.8	68.4
	Att	97.5	99.1	99.5	99.2	0.99	98.1	99.3

[1] ORG means the data set of target organ.
[2] CLF means classifier. Integra means integrative model. Att means Att-RethinkNet.

https://doi.org/10.1371/journal.pcbi.1010402.t010

Table 11. ACC_{lab} of the integrative model and Att-RethinkNet for liver data.

CLF[1]	CI	EC	HY	IM	NL	MI	NE	HN	KCP	SCN	SW	CV
Integrative model	83.3	95.8	90.1	98.5	88.2	86.6	91.9	90.0	95.0	97.9	95.0	98.7
Att-RethinkNet	96.2	99.0	98.3	98.0	97.6	97.6	96.1	98.6	97.4	97.9	97.7	98.9

[1] CLF means classifier. Columns 2 to 13 indicate the ACC_{lab} (%) of the corresponding pathological finding.

https://doi.org/10.1371/journal.pcbi.1010402.t011

that although the integrative model has obtained considerably high classification accuracy when classifying each label (although lower than the proposed method, shown in Tables 11 and 12), the subset accuracy of the integrative model is unsatisfactory (Table 10). The low subset accuracy is due to the fact that the integrative model makes predictions for each label separately, which cannot guarantee the prediction result for each pathology correct at the same time.

In terms of predicting each label, we specifically show the ACC_{lab} of the proposed model and the integrative model in Fig 12. The results prove that the proposed model has a significant improvement in correctly predicting each pathology when compared with the integrative method. The difference of ACC_{lab} between our method and the integrative method ranges from around 1% to around 12% for drug-induced liver toxicity except increased mitosis(IM) and single cell necrosis(SCN), while the ACC_{lab} difference ranges from 20% to 42% for drug-induced kidney toxicity.

The ROC curves of Att-RethinkNet and the integrative model can be found in Fig 13. From the experimental results, we can find that the curve of Att-RethinkNet lies far above that of the integrative model. The AUC of the integrative model is approximately half of that of the Att-RethinkNet.

Validation on rat liver *in vitro* data

In order to further verify the reliability of the prediction model, we carried out experiments on independent and invisible data, the *in vitro* toxicity data of rat liver. We divided the data into two parts. One part of data was used as the training set and it was augmented and balanced,

Table 12. ACC_{lab} of the integrative model and Att-RethinkNet for kidney data.

CLF[1]	HC	LCI	BC	CY	DI	CD	NE	RE
Integrative model	56.8	68.0	79.1	64.7	73.4	62.2	76.8	66.0
Att-RethinkNet	99.4	99.4	98.9	99.1	99.5	99.3	99.7	99.1

[1] CLF represents classifier. Columns 2 to 9 indicate the ACC_{lab} (%) of the corresponding pathological finding.

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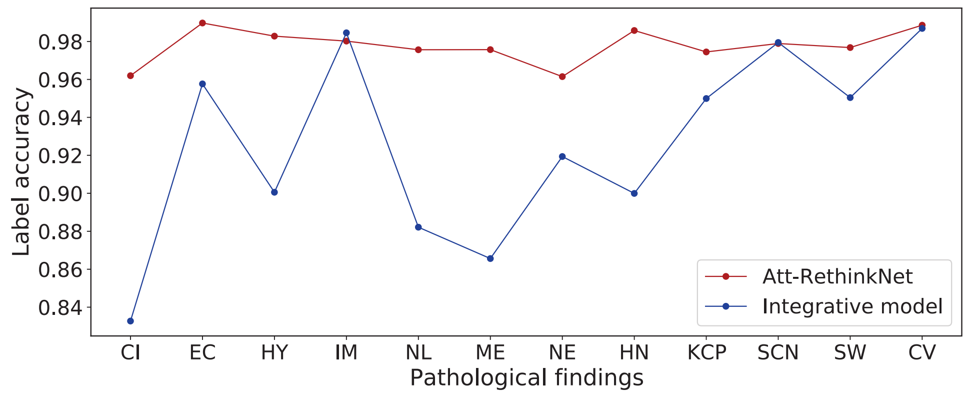
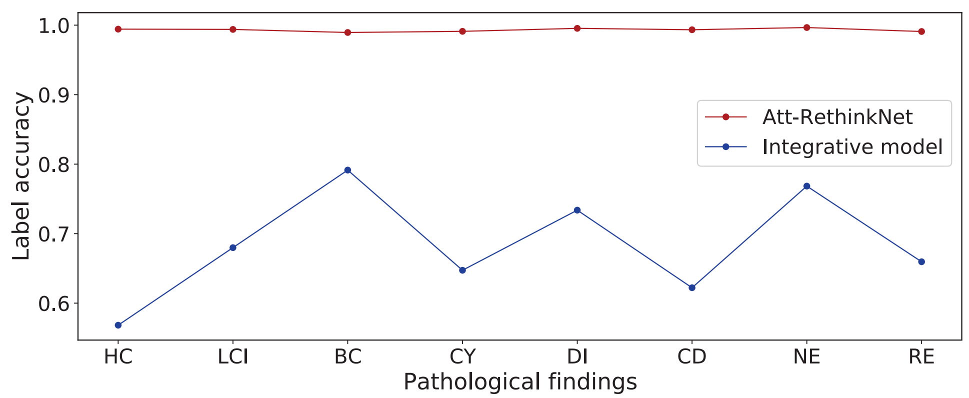


(a) 肝脏



(b) 肾

图12. 使用肝脏样本(a)和肾脏样本(b)对所提出模型与综合模型的ACC_{lab} 比较。

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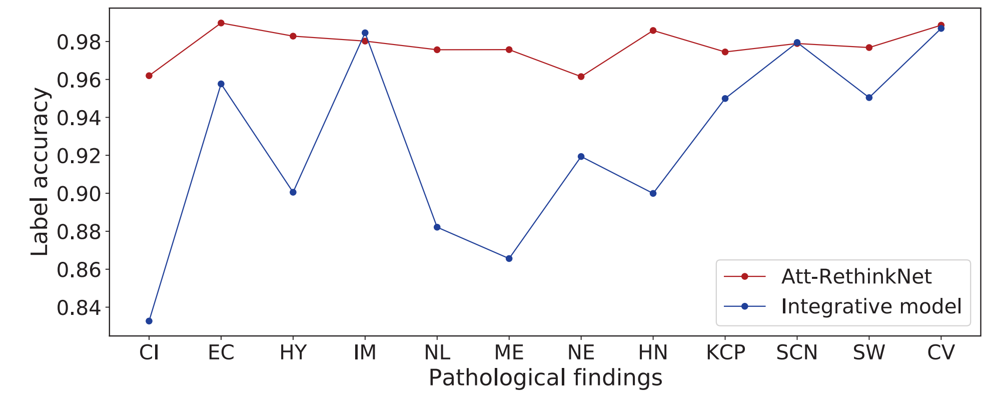
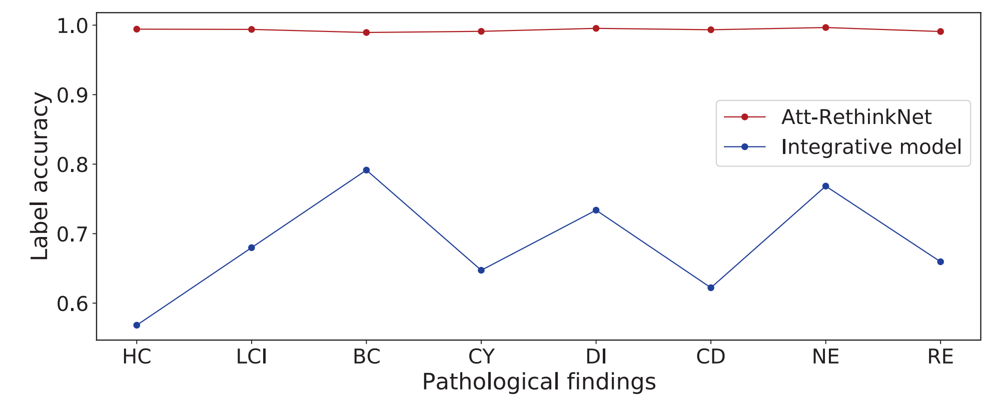
剩余的数据被用作测试集。我们选择了单次给药后24小时体外大鼠肝脏的相应基因表达水平。对这些数据进行了包括剂量-反应曲线拟合在内的预处理操作。

表 13 列出了肝脏体外数据集的分类结果。对于肝脏病理，我们基于 LSTM 的 Att-RethinkNet 模型取得了相对较高的准确率，达到了 73.0%。SEN 和 SPE 均在 80% 以上。其中 SPE 达到 94.8%，比 SEN 高出 10%。因此，我们的方法可以用于毒性评估的第一步。通过预测药物未出现任何病理性表现，可以高精度地判断药物的安全性。然而，如果结果显示药物可能引发某种病理性表现，仍可能需要进一步的安全性筛查。实验结果表明，我们提出的模型适用于新药，并能够减少过拟合，同时对不可见的测试数据进行预测。

表14进一步显示了模型在每个标签上的表现。可以清楚地看到，我们提出的模型具有预测与药物性肝损伤对应的特定病理发现的能力，并且在预测病理发现肝隔膜结节方面能够达到较高的准确性，而病理的预测能力



(a) Liver



(b) Kidney

Fig 12. ACC_{lab} comparison of the proposed model with the integrative model using liver samples (a) and kidney samples (b).

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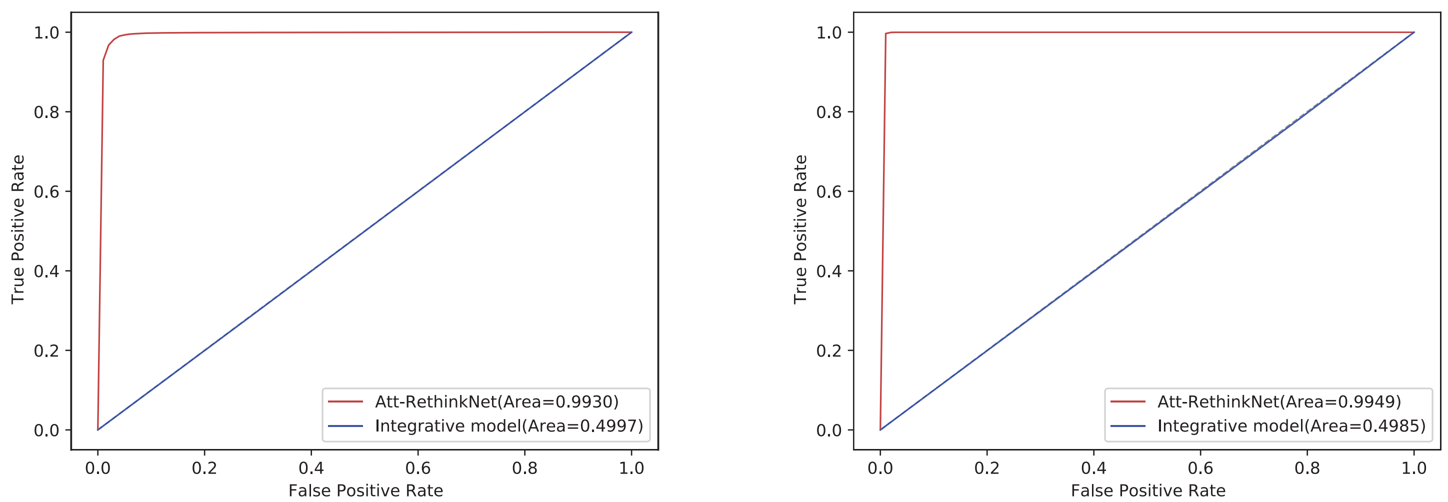
and the remaining data was used as the test set. We selected the corresponding gene expression levels of rat liver *in vitro* at the time point of 24 hours after a single dose administration. Pre-processing operations including dose-response curve fitting was operated on this data.

Table 13 lists the classification results on liver *in vitro* data set. For liver pathology, our Att-RethinkNet model based on LSTM achieved relatively high accuracy, with a value of 73.0%. The SEN and SPE are all above 80%. And the SPE achieves 94.8% which is 10% higher than the SEN. Therefore, our approach can be used in the very first step of toxicity evaluation. Drugs are determined safe with high accuracy by predicting them without any pathological findings. However, if the results show that the drug can induce a certain pathological finding, further safety screening may still be needed. The experimental results show that our proposed model is applicable to new drugs and it is able to reduce the over-fitting, and make predictions for the invisible test data.

Table 14 further shows the performance of the model on each label. It can be clearly seen that our proposed model has the ability to predict the specific pathological findings corresponding to 12 drug-induced liver toxicity, and can achieve high accuracy in predicting the pathology finding hepatodiaphragmatic nodule, while the prediction ability of pathology



(a) Liver (b) Kidney

图13. Att-RethinkNet 的 ROC 曲线 a 以及整合方法。(a) 是针对肝脏数据，(b) 是针对 肾脏数据。

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表13. 肝脏体外数据集的分类结果。

ORG[1]	CLF[2]	准确率 (%)	SEN (%)	SPE (%)	F1 (%)	AUC	ACC_{pair}(%)	ACC_{avelab}(%)
肝脏	Att-RethinkNet	73.0	84.3	94.8	84.8	0.90	80.1	89.5

[1] ORG 意味着目标器官的数据集。[2]
CLF 意味着分类器。

https://doi.org/10.1371/journal.pcbi.1010402.t013

发现坏死需要改进。总体而言，Att-RethinkNet 可以为药物开发过程提供辅助功能。

结果的ROC曲线如图14所示。从ROC曲线可以看出，所提出模型的曲线下面积为0.9，这表明该模型能够在不可见和独立数据上预测病理发现。

总之，我们提出的模型具有通用性，能够预测未知药物的毒性病理学发现。上述交叉验证实验中每个折的子集准确性及预测结果的标准差列于S2表，所提出的深度神经网络模型的参数设置列于S3表。

结论与讨论

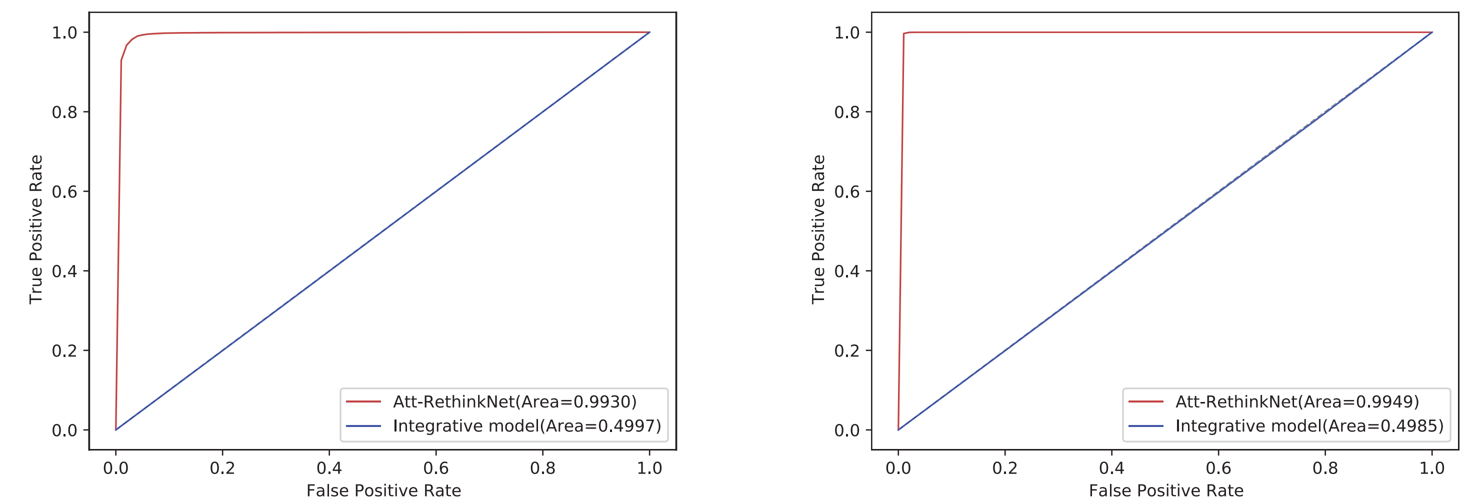
我们提出的Att-RethinkNet框架可以基于毒理基因组学数据，在预测药物引起的多器官病理方面取得优异的性能，这对于

表14. Att-RethinkNet在肝脏体外数据上的ACC_{lab}。

CLF[1]	CI	EC	HY	IM	NL	MI	NE	HN	KCP	SCN	SW	CV
Att-RethinkNet	84.1	92.9	85.1	94.7	91.2	89.9	69.7	99.0	88.4	91.0	94.8	92.9

[1] CLF 意味着分类器。第2列到第13列表示相应病理发现的 ACC_{lab} (%)。

https://doi.org/10.1371/journal.pcbi.1010402.t014



(a) Liver (b) Kidney

Fig 13. The ROC curves of the Att-RethinkNet and the integrative method. (a) is for liver data and (b) is for kidney data.

https://doi.org/10.1371/journal.pcbi.1010402.g013

Table 13. Classification results on the liver *in vitro* data set.

ORG[1]	CLF[2]	ACC (%)	SEN (%)	SPE (%)	F1 (%)	AUC	ACC_{pair}(%)	ACC_{avelab}(%)
Liver	Att-RethinkNet	73.0	84.3	94.8	84.8	0.90	80.1	89.5

[1] ORG means the dataset of target organ.
[2] CLF means classifier.

https://doi.org/10.1371/journal.pcbi.1010402.t013

finding necrosis needs to be improved. In general, Att-RethinkNet can provide auxiliary functions for the process of drug development.

The ROCs of the results are shown in Fig 14. It can be seen from the ROCs that the area under the curve of the proposed model is 0.9, which shows that the model can predict the pathological findings on invisible and independent data.

In conclusion, our proposed model is generalized, which is able to predict toxic pathological findings of unknown drugs. The subset accuracy of each fold and standard deviation of prediction results corresponding to the above cross-validation experiments are put in S2 Table, and the parameter settings of the proposed deep neural network model are put in S3 Table.

Conclusion and discussion

Our proposed Att-RethinkNet framework can achieve excellent performance in predicting drug-induced pathology in multiple organs based on toxicogenomics data, which is helpful to

Table 14. ACC_{lab} of Att-RethinkNet for liver *in vitro* data.

CLF[1]	CI	EC	HY	IM	NL	MI	NE	HN	KCP	SCN	SW	CV
Att-RethinkNet	84.1	92.9	85.1	94.7	91.2	89.9	69.7	99.0	88.4	91.0	94.8	92.9

[1] CLF means classifier. Columns 2 to 13 indicate the ACC_{lab} (%) of the corresponding pathological finding.

https://doi.org/10.1371/journal.pcbi.1010402.t014

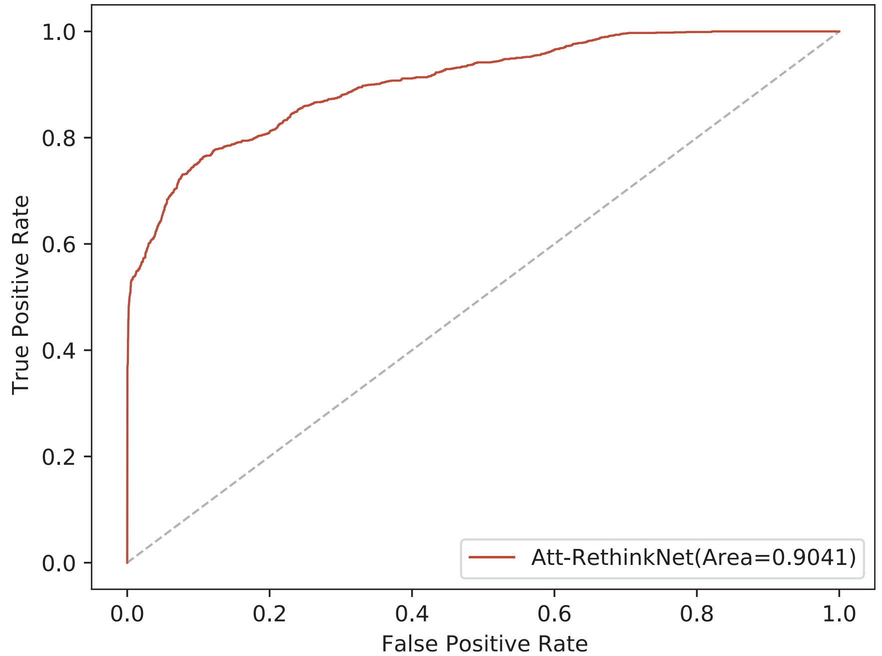


图14. Att-RethinkNet在体外肝脏数据的ROC曲线。

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在早期阶段检测和诊断器官特异性毒性，并为制药行业提供具有成本效益的解决方案。

我们的研究提出了一种基于注意力机制的RethinkNet框架，用于基于基因表达谱的药物诱导病理发现预测。该模型模拟了人类的反思过程，LSTM层之前的注意力机制关注更重要的特征。我们的模型在肝脏和肾脏上均表现出令人印象深刻的性能。所提出的Att-RethinkNet模型在体内肝脏数据上的准确率为89.4%，在体内肾脏数据上的准确率为97.5%，高于BR、CC、RethinkNet和“整合模型”。Att-RethinkNet的运行时间仅需数小时，比传统方法BR和CC快得多，后者几乎需要一周时间。

在当前的工作中，我们发现分类结果已经足够好，能够显示所提出的Att-RethinkNet 的有效性。然而，我们的实验仍然存在一些限制。例如，病理发现依赖于人工标注。当标签收集量大或药物数量多时，每个实例对应的标签定义可能会花费大量时间。此外，由于获取基因表达数据和病理信息的困难，我们仅针对肝脏和肾脏，而未涉及其他目标器官，因此在我们的研究中未对其他类型的器官进行验证。

对于我们提出的深度神经网络模型，仍有改进的空间和扩展的可能性。下一步工作，我们将考虑基因选择的分析，我们的网络模型可以优化以识别与潜在毒性高度相关的基因。实际上，我们的模型不仅限于肝脏和肾脏，也可以轻松扩展到其他器官。未来，我们的目标是构建一个更通用的模型，适用于所有器官，并整合多组学数据。

此外，所提出的方法为多标签分类问题提供了一个新的、有趣的见解，适用于多个领域，例如声音分类、图像分类和文本分类。未来的一个有趣的研究工作是探索我们多标签分类模型在……领域的应用范围。

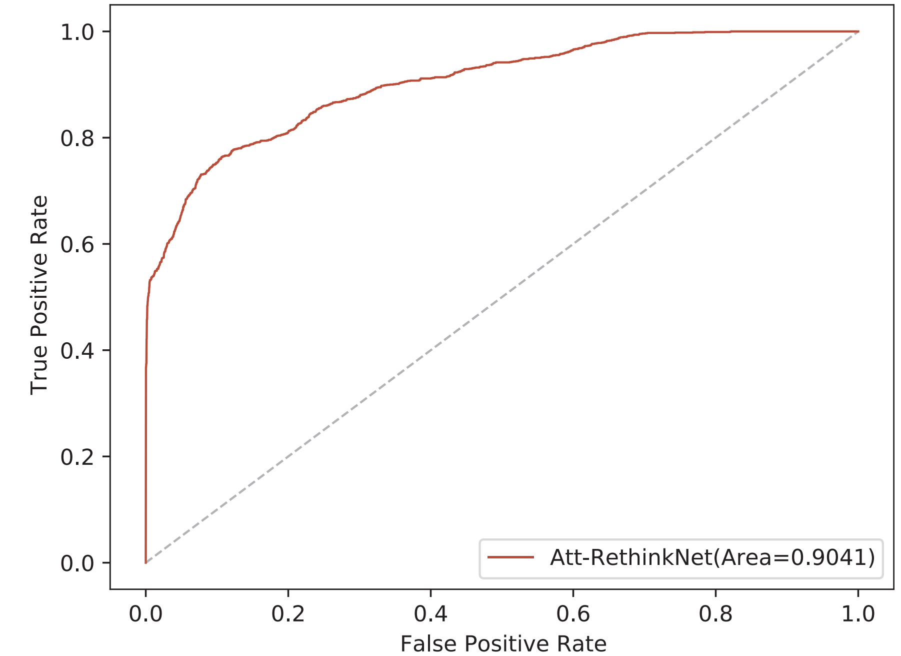

Fig 14. The ROC curves of the Att-RethinkNet for liver *in vitro* data.

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detect and diagnose organ-specific toxicity in early stage, and provides a cost effective solution for drug industry.

Our study proposed an attention-based RethinkNet framework for drug-induced pathological finding prediction based on gene expression profile. This model mimics the human rethinking procedure, and the attention mechanism before LSTM layer focuses on more important features. Our model has shown impressive performance on both liver and kidney. The accuracy of the proposed Att-RethinkNet model is 89.4% for *in vivo* liver data and is 97.5% for *in vivo* kidney data, higher than those of the BR, CC, RethinkNet, and the "integrative model". The running time of Att-RethinkNet only takes a few hours, which is much faster than the traditional method BR and CC which takes nearly a week.

In the current work, we find that the classification results are already good enough to show the effectiveness of the proposed Att-RethinkNet. However, our experiment still has several limitations. For example, pathological findings relies on manual labeling. When the collection of labels or the number of drugs is large, the definition of labels corresponding to each instance can take a lot of time. Moreover, we only targeted on liver and kidney without additional target organs due to the difficulty in obtaining gene expression data and pathological information, so there is no verification on other kinds of organs in our study.

For our proposed deep neural network model, there is still room for improvement and scope for expansion. Next work, we will consider the analysis of gene selection [42], our network model can be optimized to identify genes highly associated with potential toxicity. In fact, our model is not limited to liver and kidney and can be easily extended to other organs. In the future, we aim to construct a more generalized model which is suitable for all organs and incorporate multi-omics data.

Additionally, the proposed method provides a new and interesting insight to multi-label classification problems, which is applicable to a spectrum of domains, such as sound classification, image classification, and text categorization. It would be an interesting future work to explore the application scope of our multi-label classification model in the field of

生物信息学，例如化合物-蛋白质相互作用的预测 [43]、人类蛋白质亚细胞定位的识别以理解蛋白质功能 [44]、基于多种风险因素的早期宫颈癌诊断 [45]。

支持信息

S1 文本。改进的 MLSMOTE（多标签合成少数类过采样技术）算法，能够有效处理多标签分类中的不平衡数据集。（PDF）

S1 表格。实验数据中涉及的药物或化学化合物。(PDF)

S2 表。所有结果的标准差。（PDF）

S3 表。所开发模型的参数设置。（PDF）

S1 图。使用 t-SNE 对肝脏的所有目标病理发现进行可视化。（PDF）

S2 图。使用 t-SNE 对肾脏的所有目标病理发现进行可视化。（PDF）

S3 图。肝脏和肾脏其他折叠的病理分类混淆矩阵。(PDF)

S4 图。病理相似性矩阵，描述每个折叠的训练集中两种病理发现的共现情况。(PDF)

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bioinformatics, such as prediction of compound-protein interactions [43], identification of human protein subcellular localization for understanding protein functions [44], diagnosing cervical cancer at early stages based on multiple risk factors [45].

Supporting information

S1 Text. The improved MLSMOTE (Multilabel Synthetic Minority Over-sampling Technique) algorithm which effectively handles the imbalanced data set for multi label classification.
(PDF)

S1 Table. The drugs or chemical compounds involved in the experimental data.
(PDF)

S2 Table. The standard deviation of all the results.
(PDF)

S3 Table. The parameter setting of the developed model.
(PDF)

S1 Fig. The visualization of all targeted pathological findings using t-SNE for liver.
(PDF)

S2 Fig. The visualization of all targeted pathological findings using t-SNE for kidney.
(PDF)

S3 Fig. The confusion matrix of the pathology classification of other folds for both liver and kidney.
(PDF)

S4 Fig. The pathology similarities matrix that describes co-occurrences of two pathological findings within training set of each fold.
(PDF)

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